README

Contents

Z80 ROMless SBC - WIP Engineering & Build Specification

Repository: github.com/gloveboxes/Z80ROMlessSBC **PDF edition:** README.pdf

Overview

This document describes a theoretical Z80 single-board computer design that has not yet been built, wired, or validated in hardware. Treat it as an engineering proposal and bring-up plan, not as a proven reference design. Every electrical assumption, timing margin, and firmware interaction still needs bench validation with the staged tests in Section 8 before the design should be considered reliable.

The proposed system is a ROMless Z80 computer supervised by a Raspberry Pi Pico 2 W. The Pico supplies the Z80 clock, holds and releases reset, loads a boot image directly into SRAM by taking the Z80 bus, and then lets the Z80 execute from RAM. A 64 KB SRAM provides the Z80 memory space; level shifters and bus transceivers isolate the Pico's 3.3 V GPIO from the 5 V Z80/SRAM bus; an MCP23S17 expands the Pico's address-drive capability during DMA and trapped I/O cycles.

The key design idea is that the Pico acts as both supervisor and virtual peripheral controller. It can stop the fully static Z80 clock on an I/O cycle, inspect the requested port, exchange one data byte, and then resume execution. Terminal I/O is intended to be one of those virtual peripherals: Z80 `IN` and `OUT` instructions feed nonblocking queues on the Pico, while a WebSocket terminal server runs on the Pico's other core so Wi-Fi and browser traffic do not disturb Z80 timing.

Z80 software lives in a reserved region of the Pico's own onboard flash rather than on removable media or compiled into the running firmware image. The Pico reads boot binaries, monitor images, and CP/M disk images directly from that memory-mapped flash region, then uses the already-validated DMA path to populate SRAM before the Z80 is released. Once the Z80 is running, the same I/O trap mechanism can expose sector-oriented virtual disk ports backed by that same flash partition (Section 6.3).

The build plan is deliberately phased. Early phases prove power, translation, SPI expansion, bus isolation, SRAM DMA, and clock control before the Z80 is installed. Later phases prove virtual-ROM boot, synchronous I/O trapping, WebSocket terminal service, and finally the maximum qualified clock rate. Passing an earlier phase is a prerequisite for trusting the assumptions used by the next one.

Firmware and CP/M build

The repository implements the ten cumulative firmware stages under `src/`. Stage 10 includes the journaled flash disks, CP/M boot loader, and WebSocket terminal. Its host build also assembles the board-native CP/M 2.2 BIOS, reconstructs pristine CCP/BDOS bytes from the preserved Altair source image, and emits all provisionable images.

Install CMake, Ninja, Python 3, `picotool`, and `z80asm`. The firmware needs a complete Arm GNU bare-metal toolchain with Newlib; the Homebrew compiler alone does not provide the required runtime. A known working setup is:

```
brew install cmake ninja picotool z80asm
export PATH="$HOME/.local/share/arm-gnu-toolchain-15.3.rel1/bin:$PATH"
cmake -S . -B build -G Ninja \
  -DPICO_BOARD=pico2_w \
  -DZ80_WIFI_SSID='your-network' \
  -DZ80_WIFI_PASSWORD='your-password'
cmake --build build --target z80_cpm_images -j
```

Wi-Fi credentials are written only to the generated build tree. An empty SSID leaves networking disabled while flash-disk service continues to operate. The `z80_cpm_images` target builds Stage 10 and writes these files to `build/cpm/`:

Artifact	Purpose
`z80boot.pkg`	Manifest, CRCs, and the reset-ready 64 KiB Z80 image
`drive_a_cpm63k.img`	Native CP/M system disk with the SBC BIOS
`drive_b_bdsc.img` through `drive_d_blank.img`	Converted 320 KiB disks
`z80romless-flash.bin`	Complete 4 MiB initial-provisioning image
`manifest.json`	Geometry, addresses, sizes, and SHA-256 values

Run the host regression checks with:

```
python3 -m unittest discover -s src/cpm -p 'test_*.py' -v
```

PDF edition

The checked-in `README.pdf` is generated from this file. The renderer preserves tables, highlighted code, local images, links, and MathML. Each Mermaid diagram gets a dedicated page; wide diagrams use A4 landscape pages and are rendered as vector graphics without the interactive preview controls.

Install Node.js, Pandoc, and a Chromium-family browser, then run:

```
brew install node pandoc
npm ci
npm run pdf
```

The script searches for Microsoft Edge, Google Chrome, or Chromium. Set `BROWSER_EXECUTABLE` to use another installation. Optional input and output paths may be passed after `--`, for example:

```
npm run pdf -- README.md README.pdf
```

Flash provisioning

For a new or disposable flash, put the Pico in BOOTSEL mode and write the complete image. This erases existing CP/M disk contents and the journal:

```
picotool load -v build/cpm/z80romless-flash.bin -t bin -o 0x10000000
picotool reboot
```

For normal updates, load only the firmware and selected storage regions. These commands preserve regions not named by the input file:

```
picotool load -v build/src/stage10_websocket_terminal/z80_stage10_websocket_terminal.uf2
picotool load -v build/cpm/z80boot.pkg -t bin -o 0x102A0000
picotool load -v build/cpm/drive_a_cpm63k.img -t bin -o 0x102C0000
picotool load -v build/cpm/drive_b_bdsc.img -t bin -o 0x10310000
picotool load -v build/cpm/drive_c_escape.img -t bin -o 0x10360000
picotool load -v build/cpm/drive_d_blank.img -t bin -o 0x103B0000
picotool reboot
```

After association, browse to `http://<pico-dhcp-address>:8088/`. The USB serial console reports Stage 10 startup and accepts `s` to print terminal queue/drop counts and disk/fatal status. CP/M disk writes are persistent, so retain host backups before full reprovisioning.

0. Project Inventory

The quantities below build one complete three-breadboard prototype. Purchase quantities include a small allowance for breadboard spares.

0.1 Semiconductors

Required	Component	Package	Function
1	Raspberry Pi Pico 2 W	Module with two 20-pin headers	Supervisor, clock, DMA, virtual I/O, and Wi-Fi terminal
1	Z84C0020PEC	40-pin PDIP	CMOS Z80 CPU
1	AS6C1008-55PCN	32-pin PDIP	SRAM; lower 64 KB used
1	MCP23S17-E/SP	28-pin SPDIP	SPI-to-16-bit address interface
3	SN74AHCT125N	14-pin PDIP	Pico 3.3 V-to-5 V control and SPI translation
2	SN74HCT245N	20-pin PDIP	Bidirectional address-bus isolation
1	SN74LVC245AN	20-pin PDIP	Pico data-bus interface
1	SN74LVC244AN	20-pin PDIP	5 V-to-3.3 V buffering for Z80 status/control and MCP SO
1	74HCT157	16-pin PDIP	SRAM CE#/OE#/WE# source arbitration
1	74HCT08	14-pin PDIP	AND-gates RESET# and BUSACK# to drive the 74HCT157 select line
1	1N5819	Axial diode	Schottky power OR from external +5 V to Pico VSYS

> **New required additions:** The original design required SRAM control arbitration but did not assign a part; the 74HCT157 implements that requirement (Section 1.2). The design also needs a Pico-driven SRAM CE# line and full 5 V translation of the MCP23S17 SPI inputs, which together need 9 translated signals (RESET#, BUSREQ#, CLK, SRAM WE#, SRAM OE#, SRAM CE#, SPI CS#, SPI SCK, SPI SI) and do not fit in the original two 74AHCT125N packages (8 gates). A third SN74AHCT125N (IC3) is added; see Section 4. Selecting the 74HCT157 purely from BUSACK# also leaves SRAM control unreachable by the Pico whenever RESET# is held (Section 8.9) or the Z80 is physically absent from its socket (Section 8.7); a 74HCT08 AND gate combines RESET# and BUSACK# ahead of the select input to fix both cases (Section 1.2). The SN74LVC244AN buffers RD#/WR# because Pico 2 GP27 and GP28 are standard ADC-capable pads, not 5 V-tolerant FT pads. It also buffers BUSACK#, IORQ#, and MCP SO: although GP0, GP1, and GP20 are FT pads, their 5.5 V tolerance requires RP2350 IOVDD to be present at 3.3 V. Using the spare LVC244 channels preserves safe power sequencing and powered-off isolation (Section 5.3).

0.2 Sockets and Headers

Quantity	Item
1	40-pin DIP socket
1	32-pin DIP socket
1	28-pin DIP socket
4	20-pin DIP sockets
1	16-pin DIP socket
4	14-pin DIP sockets
2	20-pin 0.1-inch male headers for the Pico 2, if not fitted
2	20-position 0.1-inch socket strips for removable Pico mounting; do not substitute DIP IC sockets

0.3 Capacitors and Resistors

Fitted	Purchase	Item	Purpose
12	15	100 nF X7R ceramic capacitors, at least 10 V	One at every DIP IC supply pair
3	5	10 uF capacitors, at least 10 V	One per breadboard
1	2	47 uF electrolytic capacitor, at least 10 V	5 V supply-entry bulk capacitance
24	30	10 kOhm, 1/4 W resistors	Defined startup levels and temporary test pulls
Reused for tests	10	1 kOhm, 1/4 W resistors	First-drive current limiting and manual input tests

Place each 100 nF capacitor directly across its IC supply pins with the shortest practical leads. On the 5 V side, fit 10 kOhm pull-ups to RESET#, BUSREQ#, BUSACK#, MREQ#, IORQ#, RD#, WR#, MCP SO, WAIT#, INT#, and NMI#. On the Pico side, use 10 kOhm pull-ups to 3.3 V on GP4 (BUSREQ#), GP5 (SRAM CE#), GP7/GP9 (transceiver OE#), GP21 (SPI CS#), GP22 (SRAM WE#), and GP26 (SRAM OE#). Use 10 kOhm pull-downs to GND on GP2 (CLK), GP3 (RESET#), GP6/GP8 (transceiver DIR), and GP18/GP19 (SPI SCK/SI). These external resistors establish safe levels before the firmware configures SIO; the 5 V pull-ups alone cannot override an enabled AHCT output. Tie every unused CMOS input to a defined level.

0.4 Construction and Power

Quantity	Item	Requirement
3	830-tie-point solderless breadboards (BusBoard/X-ON BB830, ABS body; 63 terminal rows plus two 100-point power-distribution strips per board)	Memory, CPU core, and peripheral zones
1	Regulated 5 V supply	At least 1 A; adjustable current limiting preferred
1	USB data cable	Pico programming and serial diagnostics
1 set	22 AWG solid-core hookup wire	Multiple colors for signal groups
1 set	Short male-to-male jumpers	Inter-board and temporary connections
1 set	Short ground and power-distribution links	Connects all three boards; includes +5 V, 3.3 V, and multiple GND links
1	In-line power switch or removable supply jumper	Emergency power removal; rated at least 1.5 A continuous
As required	Labels or wire markers	Bus bits, controls, and socket pin 1

Feed the breadboard's +5 V logic rail directly from the regulated supply. Feed Pico VSYS from that rail only through the 1N5819: anode to external +5 V, banded cathode to VSYS. Pico 2 already has a Schottky diode from USB VBUS to VSYS, so this second diode safely ORs USB and external power without back-powering either source. Never link the external +5 V rail directly to Pico VBUS or VSYS. Feed the SN74LVC245AN and SN74LVC244AN only from the Pico 3.3 V rail; all other logic uses the regulated 5 V rail. No 5 V output connects directly to a Pico GPIO; the SN74LVC244AN's `Ioff` protection keeps the five monitored inputs isolated while its 3.3 V supply is absent or ramping. Tie the Pico's AGND pin (pin 33) to the common digital ground: this design does not use the ADC function on GP26-GP29, so no separate analogue ground plane is needed.

0.5 Bring-Up Equipment

- Digital multimeter with resistance, continuity, and DC voltage modes.
- Two-channel oscilloscope rated for at least 100 MHz for edge-integrity checks; a 20 MHz instrument is adequate only for low-speed functional bring-up.
- Logic analyzer with at least 16 channels; 24 or more is preferred.
- Current-limited bench supply or an in-line 5 V current meter.
- Fine probe hooks or test clips suitable for DIP pins.

0.6 Optional Items

- LEDs with 1 kOhm series resistors for low-speed diagnostics only; do not permanently load high-speed buses with LEDs.
- No card socket or extra storage hardware is needed for boot images or CP/M disks: they live in a reserved region of the Pico's own onboard flash (Section 6.3), provisioned once with `picotool` and read back over the existing memory-mapped XIP bus. Reflash that region with `picotool` to change a disk image; no wiring changes.
- Spare 100 nF capacitors, resistors, jumper wire, and an IC extractor.

1. Reference Pin Mapping & Logic Domain Verification

This architecture leverages the fully static nature of the Z84C0020PEC CMOS CPU. The Pico controls the clock and uses the Z80 BUSREQ#/ BUSACK# handshake for DMA ownership. During RESET, the address and data buses float and control outputs are inactive. During BUSACK#, A0-A15, D0-D7, MREQ#, IORQ#, RD#, and WR# float while BUSACK# remains actively driven LOW; M1#, RFSH#, and HALT# are not part of the floated bus set.

1.0 Raspberry Pi Pico 2 Header Pin Map

GPIO numbers in the firmware are not physical header numbers. Use this table for construction; the ranges preserve ascending bit order.

Pico function	GPIO	Physical header pin	Connection
BUSACK# monitor	GP0	1	SN74LVC244AN 1Y1
IORQ# trap input	GP1	2	SN74LVC244AN 1Y2
Z80 CLK	GP2	4	74AHCT125 IC1 Gate 3 input
Z80 RESET#	GP3	5	74AHCT125 IC1 Gate 1 input
Z80 BUSREQ#	GP4	6	74AHCT125 IC1 Gate 2 input
SRAM CE#	GP5	7	74AHCT125 IC2 Gate 2 input
Data DIR / OE#	GP6 / GP7	9 / 10	SN74LVC245 pins 1 / 19
Address DIR / OE#	GP8 / GP9	11 / 12	Both SN74HCT245 pins 1 / 19
D0-D3	GP10-GP13	14-17	SN74LVC245 B1-B4
D4-D7	GP14-GP17	19-22	SN74LVC245 B5-B8
SPI SCK / MOSI / MISO / CS#	GP18-GP21	24-27	IC3 Gates 2/3, LVC244 2Y1, IC3 Gate 1
SRAM WE#	GP22	29	74AHCT125 IC1 Gate 4 input
SRAM OE#	GP26	31	74AHCT125 IC2 Gate 1 input
Z80 RD# monitor	GP27	32	SN74LVC244AN 1Y3
Z80 WR# monitor	GP28	34	SN74LVC244AN 1Y4
3.3 V output	-	36	SN74LVC245/LVC244 VCC and 3.3 V pull-ups
VSYS	-	39	1N5819 banded cathode; anode to external +5 V

Connect Pico GND pins 3, 8, 13, 18, 23, 28, and 38, plus AGND pin 33, to common ground with multiple short links. Leave RUN pin 30, ADC_VREF pin 35, and 3V3_EN pin 37 open. Do not connect the external 5 V rail to VBUS pin 40; USB supplies that node internally.

Component	Signal Name / Group	Hardware Pin Numbers	Electrical Constraint / Logic Domain
Z84C0020PEC (CPU)	CLK	Pin 6	Input. Driven by level-shifted 5V CMOS square wave. Clock can be safely frozen indefinitely in either a HIGH or LOW state.
	IORQ#	Pin 20	Output. Active Low. Buffered to 3.3 V through SN74LVC244AN channel 1A2/1Y2 and monitored by Pico 2 GP1 to trigger clock-stop trapping.
	MREQ#	Pin 19	Output. Active Low, three-state. Connects to 74HCT157 channel 3B for CPU-owned SRAM cycles; pulled HIGH when the CPU is absent.
	RD#	Pin 21	Output. Active Low. Buffered to 3.3 V through SN74LVC244AN channel 1A3/1Y3 and sampled by Pico 2 GP27 to resolve cycle intent.
	WR#	Pin 22	Output. Active Low. Buffered to 3.3 V through SN74LVC244AN channel 1A4/1Y4 and sampled by Pico 2 GP28 to resolve cycle intent alongside RD#.
	BUSACK#	Pin 23	Output. Active Low. Feeds 74HCT157 arbitration via the 74HCT08 AND gate (Section 1.2) and is buffered to 3.3 V through SN74LVC244AN channel 1A1/1Y1 for monitoring at Pico 2 GP0.
	BUSREQ#	Pin 25	Input. Active Low. Level-shifted up to 5V to request DMA ownership.
	RESET#	Pin 26	Input. Active Low. Driven by 74AHCT125 IC1 Gate 1 and also feeds 74HCT08 Gate 1 for SRAM-source arbitration. Hold LOW for at least three full clock cycles.
	M1#	Pin 27	Output. Active Low. Probe point for the Phase 7 opcode-fetch test; not connected to Pico logic.
	A0 – A15 (Address Bus)	Pins 30 – 40, 1 – 5	5V Tri-state Bus. Driven by CPU during active run, floats during DMA/RESET.
	D0 – D7 (Data Bus)	Pins 7 – 10, 12 – 15	5V Bi-directional Tri-state Bus. Connected directly to SRAM data bus pins.
	WAIT#	Pin 24	Input. Active Low. Unused; pull to +5V through 10 kOhm so it never floats LOW and stalls the CPU in a permanent wait state.
	INT#	Pin 16	Input. Active Low. Unused; pull to +5V through 10 kOhm to prevent a spurious interrupt from a floating input.
	NMI#	Pin 17	Input. Active Low, edge-sensed. Unused; pull to +5V through 10 kOhm to prevent a spurious non-maskable interrupt from a floating input.

Component	Signal Name / Group	Hardware Pin Numbers	Electrical Constraint / Logic Domain
AS6C1008-55PCN (SRAM)	A0 – A15 (Address lines)	Pins 12-10, 9-7, 6-5, 27-26, 23, 25, 4, 28, 3, 31	5V Input. Connected directly to the 5V Z80 address bus. Driven by MCP23S17 during DMA.
	A16	Pin 2	5V Input. Tie to GND to select the lower 64KB bank; do not leave floating.
	D0 – D7 (Data lines)	Pins 13 – 15, 17 – 21	5V Bi-directional Bus. Connected directly to the Z80 data bus.
	WE#	Pin 29	Input. Active Low. Driven through the 74HCT157 arbitration mux (Section 1.2): Z80 WR# during CPU ownership, Pico DMA write signal during DMA ownership.
	OE#	Pin 24	Input. Active Low. Driven through the 74HCT157 arbitration mux (Section 1.2): Z80 RD# during CPU ownership, Pico DMA read signal during DMA ownership.
	CE#	Pin 22	Input. Active Low. Driven through the 74HCT157 arbitration mux (Section 1.2): Z80 MREQ# during CPU ownership so I/O cycles cannot access SRAM, Pico DMA chip-enable signal during DMA ownership.
	CE2	Pin 30	Input. Active High. Tie permanently to VCC (+5V) to enable the device through the active-low CE# input.

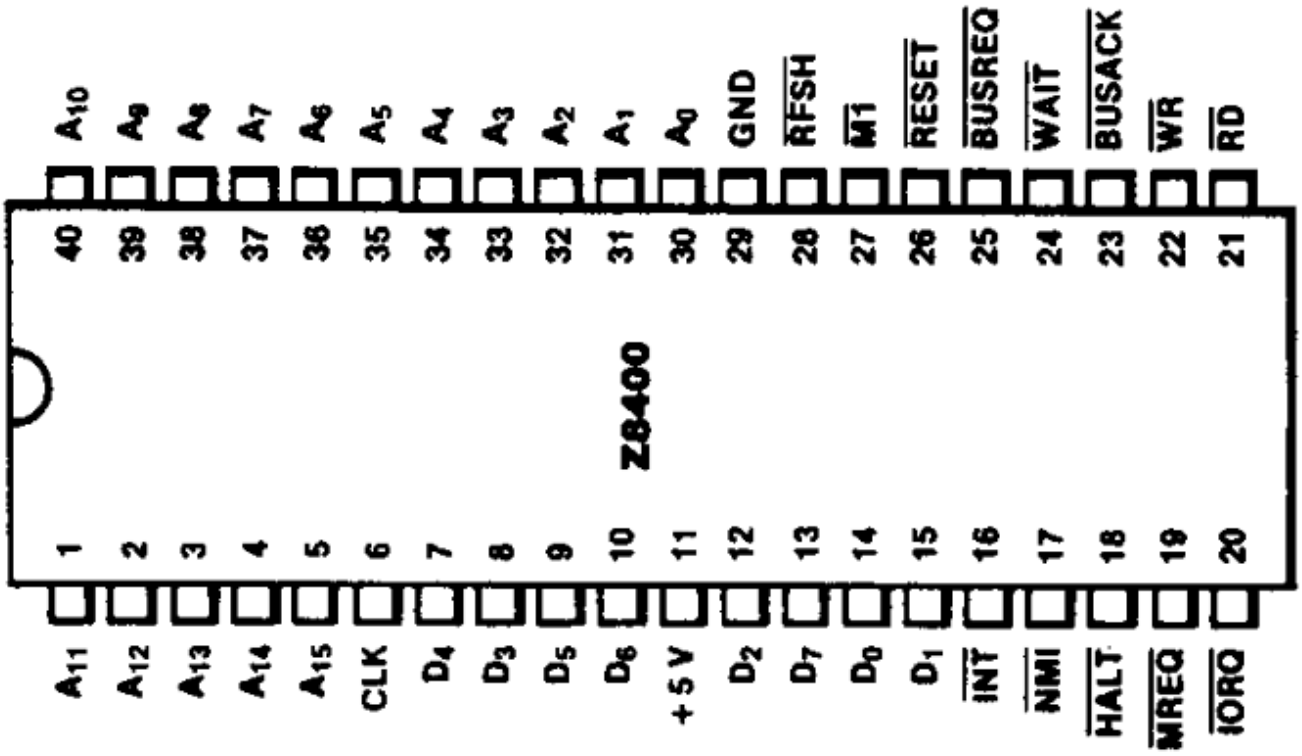
1.1 Z84C0020PEC-to-AS6C1008-55PCN Wiring

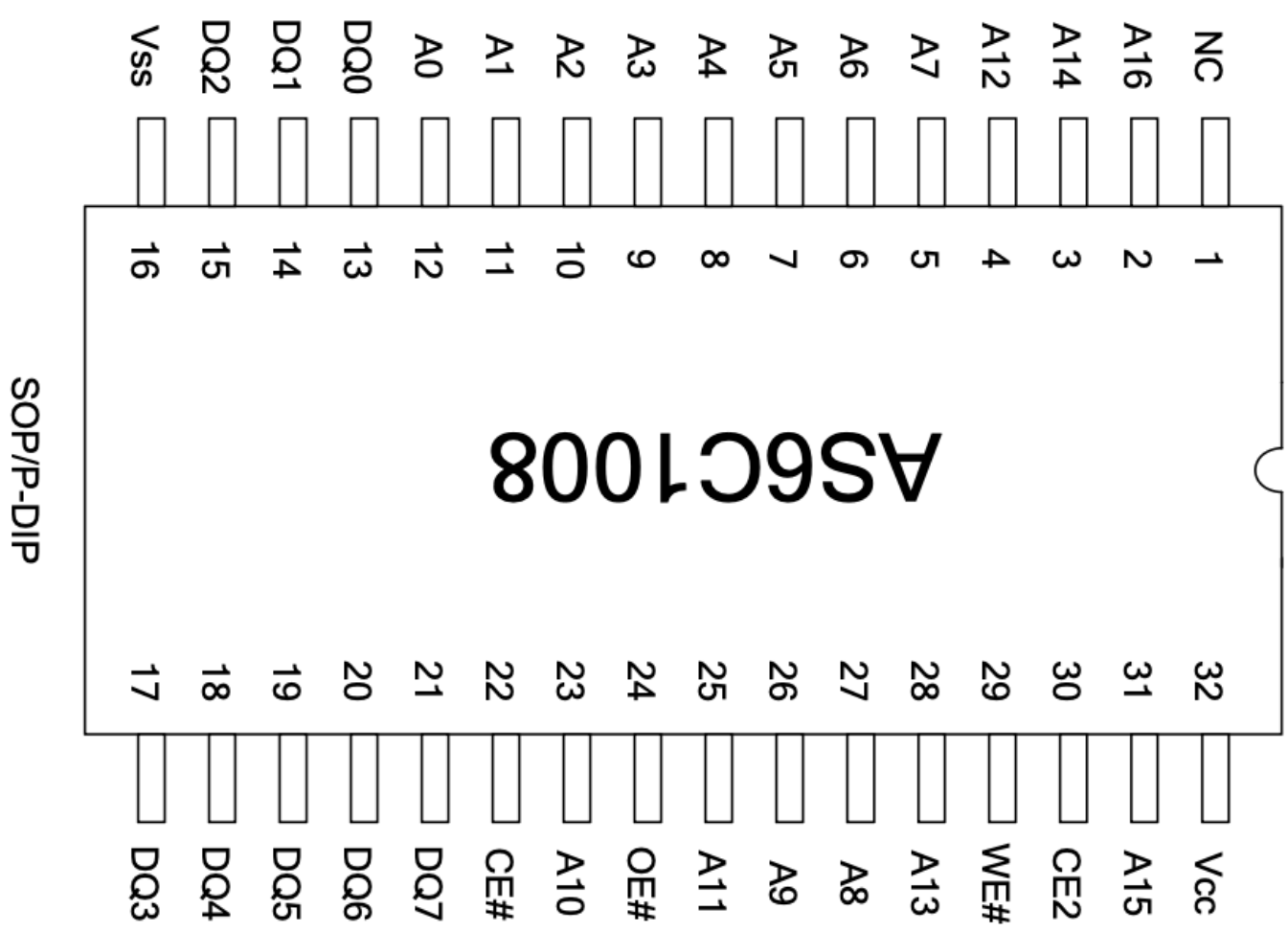
The following is the direct run-mode connection verified against the manufacturers' 40-pin PDIP and 32-pin PDIP pin assignments. The AS6C1008 is a 128K x 8 device, so A16 must be tied LOW to expose its lower 64KB as the Z80's address space. CE2 must be tied HIGH, and pin 1 is not connected.

Manufacturer datasheets:

- Zilog Z84C00 CMOS Z80 CPU Product Specification
- Alliance Memory AS6C1008 128K x 8 Low Power CMOS SRAM Datasheet

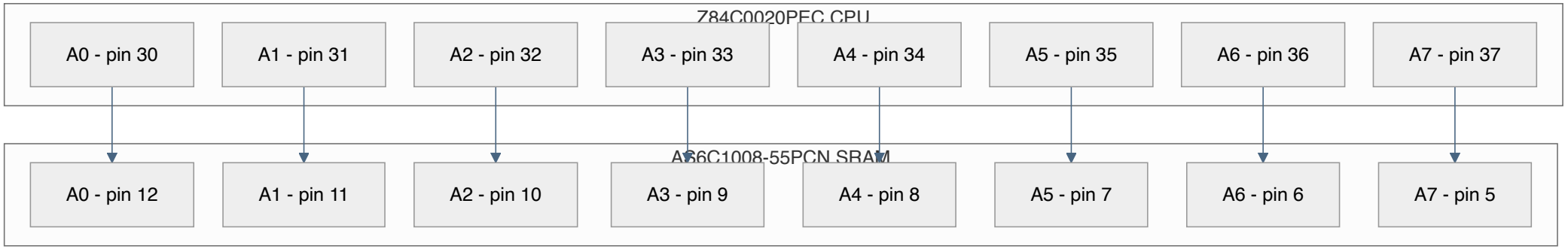
Package Pinouts





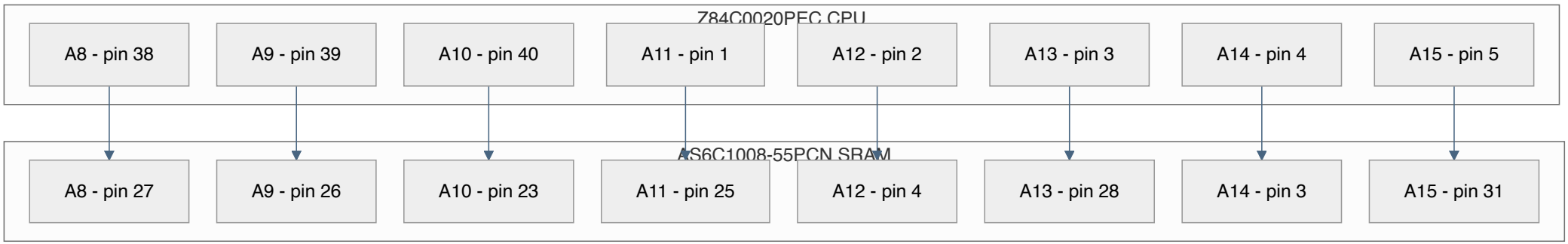
Address Bus A0-A7

Address Bus A0-A7



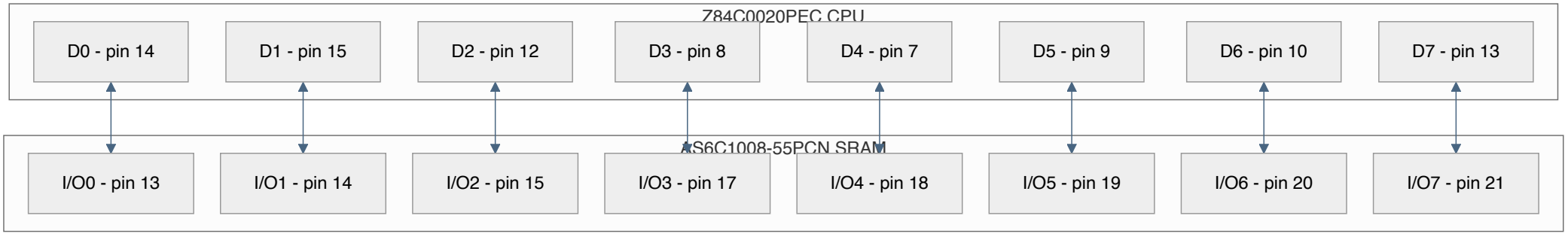
Address Bus A8-A15

Address Bus A8-A15



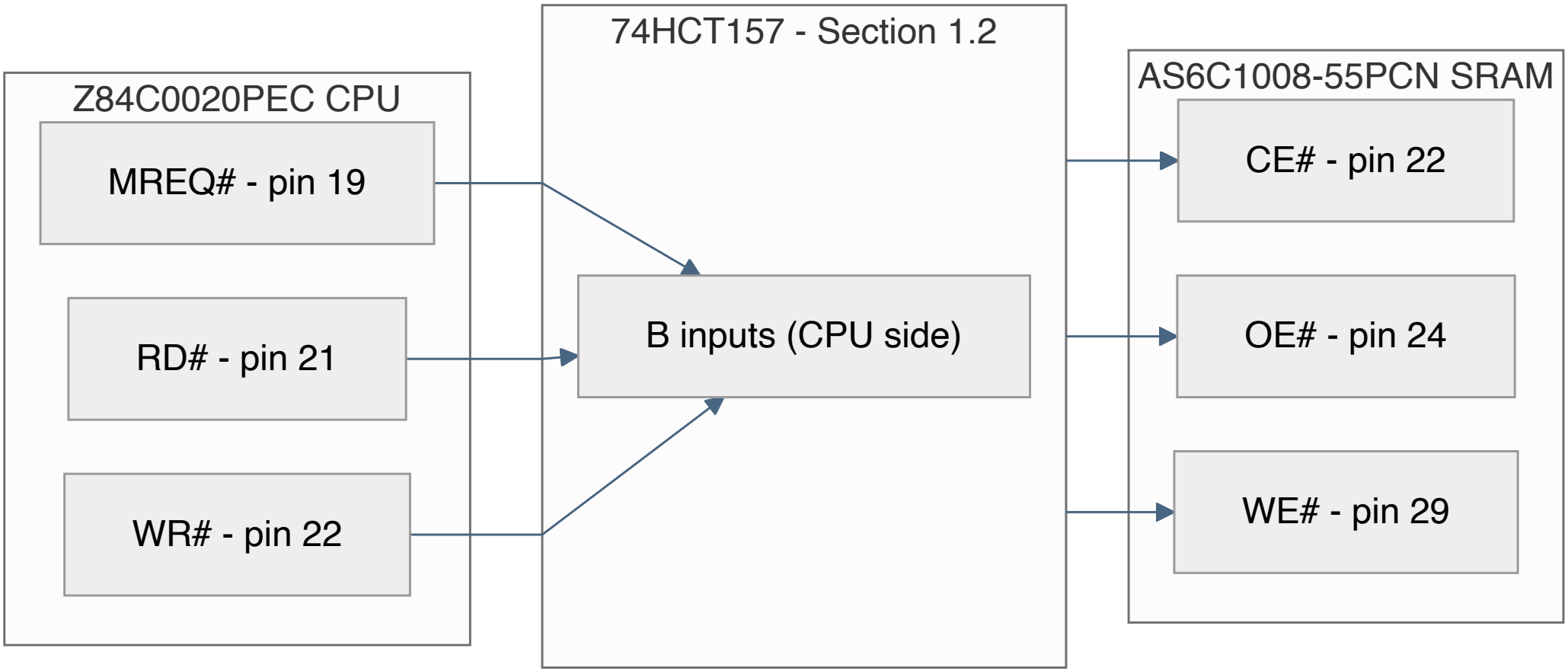
Data Bus D0-D7

Data Bus D0-D7



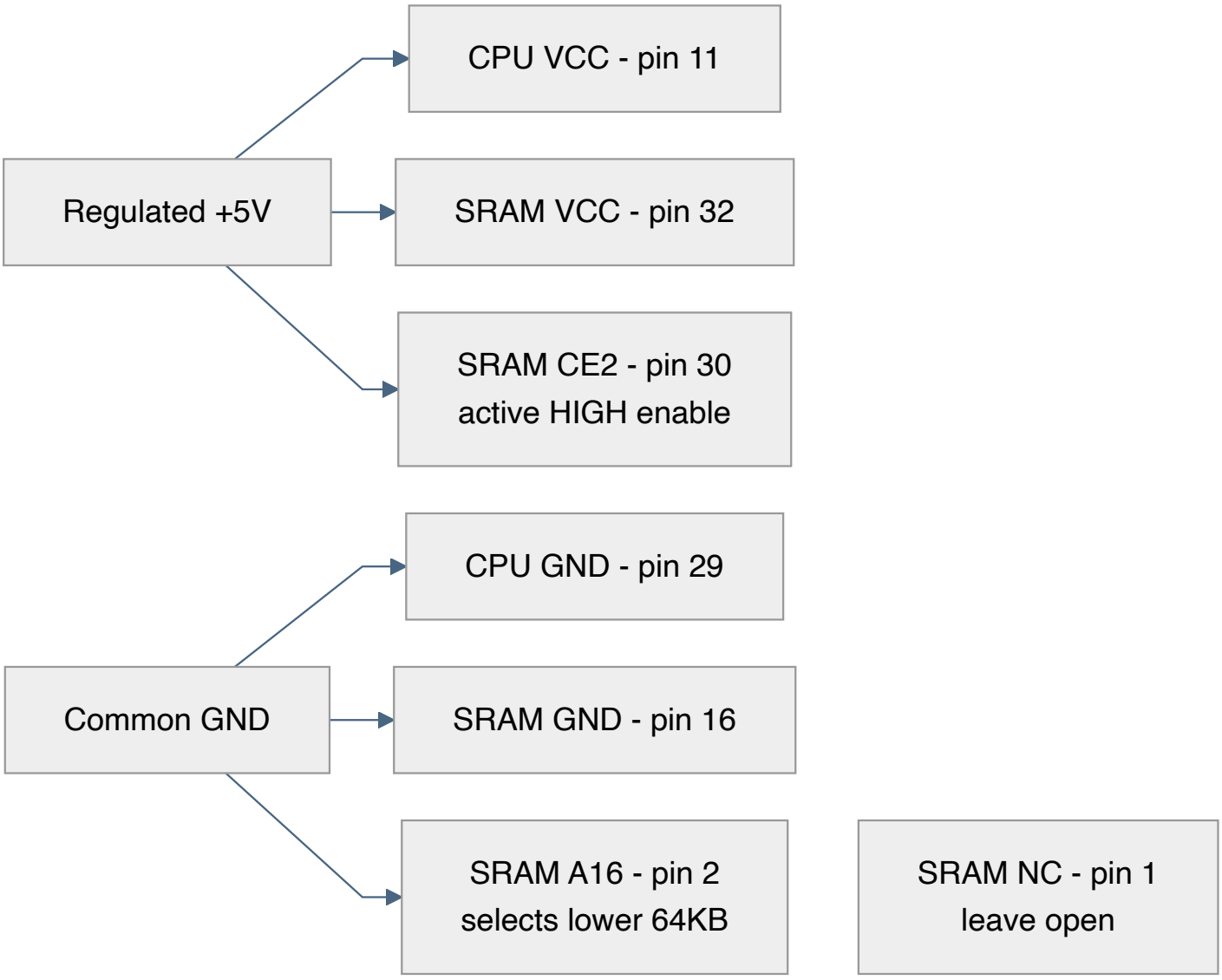
Memory Control

Memory Control



Power and Fixed Pins

Power and Fixed Pins



> **SRAM control-source arbitration:** MREQ#/RD#/WR# above are the CPU-owned run-mode paths and connect only to the 74HCT157 "B" inputs described in Section 1.2, never directly to the SRAM. The 74HCT157 selects between this path and the Pico's level-shifted DMA controls so the two sources are never joined.

1.2 SRAM Control-Source Arbitration: 74HCT157 and 74HCT08

The Z80 and the Pico must never drive SRAM CE#, OE#, or WE# at the same time. A 74HCT157 quad 2:1 multiplexer selects between the two sources. Per the SN74HCT157 datasheet function table, SELECT (pin 1, "A/B") = HIGH routes the "B" inputs to the outputs, and SELECT = LOW routes the "A" inputs; the strobe (pin 15, "G") forces every output LOW when HIGH, so it must be tied to GND to keep the mux permanently enabled.

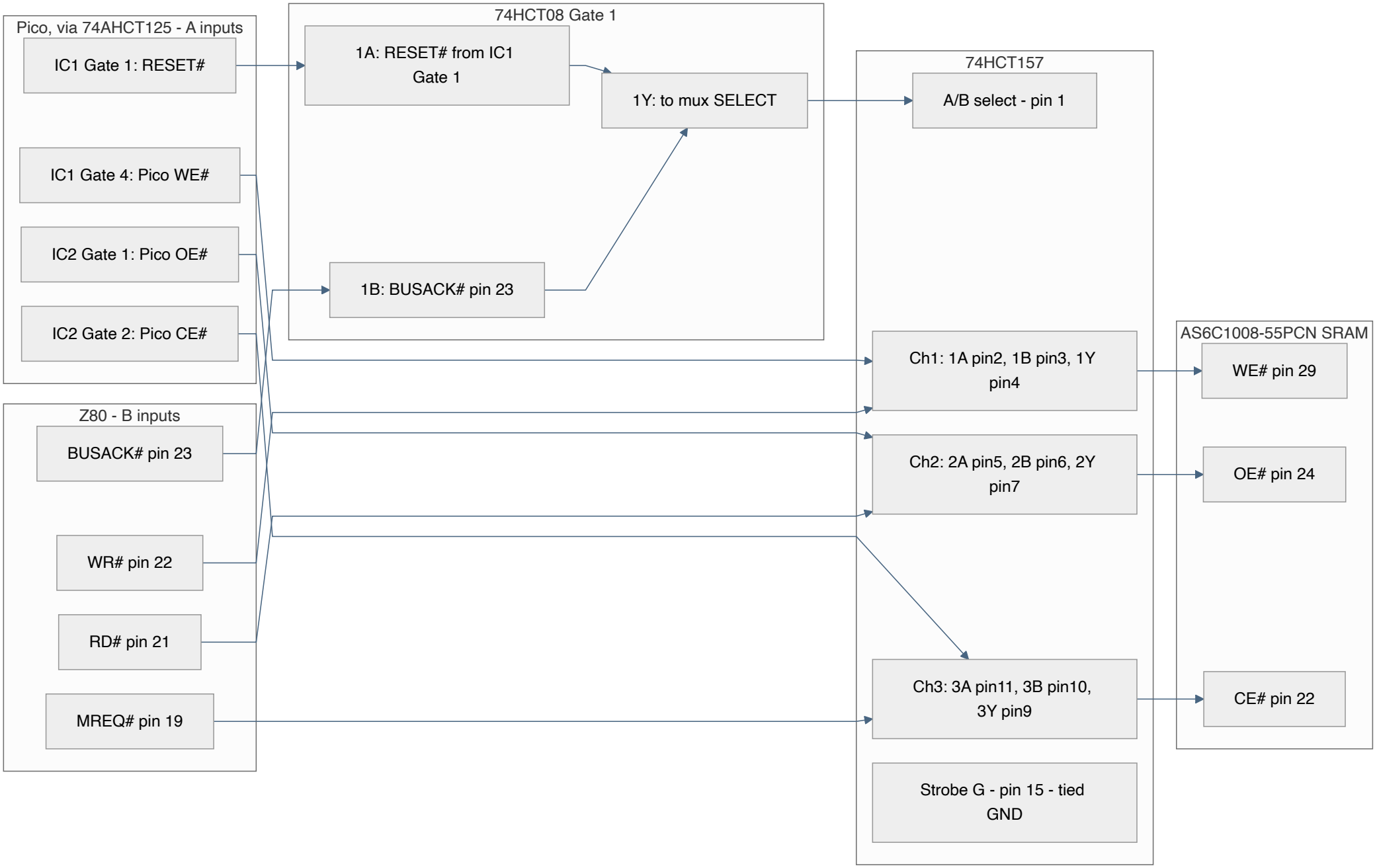
Selecting purely on the Z80's own BUSACK# output does not cover two cases this design needs. First, BUSACK# is driven to its inactive HIGH level throughout RESET, not floated (Section 1), so a held-reset DMA injection per Section 8.9 would still route SRAM control to the Z80's own inactive MREQ#/RD#/WR# lines instead of to the Pico. Second, with the Z80 physically absent from its socket, as Phase 6 (Section 8.7) requires, BUSACK# has no driver at all and only floats HIGH through its pull-up (Section 0.3), again selecting the Z80 side when the Pico needs DMA control. A single 74HCT08 AND gate resolves both cases by combining RESET# and BUSACK# ahead of the mux's select input: `SELECT = RESET# AND BUSACK#`. Because both inputs are active-low, `SELECT` is LOW (Pico "A" side) whenever *either* RESET# or BUSACK# is asserted, and HIGH (Z80 "B" side) only once the CPU is out of reset and holds the bus — exactly the three states this design needs.

- **74HCT08 Gate 1 inputs:** 1A (pin 1) from 74AHCT125 IC1 Gate 1 output (5 V RESET#, the same node driving Z80 pin 26); 1B (pin 2) from Z80 BUSACK# (pin 23), direct connection.
- **74HCT08 Gate 1 output:** 1Y (pin 3) to 74HCT157 SELECT (pin 1), replacing a direct BUSACK# connection.
- **74HCT08 unused gates:** Tie 2A/2B (pins 4-5), 3A/3B (pins 9-10), and 4A/4B (pins 12-13) to GND; leave 2Y/3Y/4Y (pins 6, 8, 11) open.
- **74HCT08 power:** VCC (pin 14) to +5 V; GND (pin 7) to ground.
- **Select (74HCT157 pin 1):** 74HCT08 Gate 1 output (above). HIGH (CPU running, out of reset) selects the Z80 "B" side; LOW (RESET# asserted or DMA active) selects the Pico "A" side.
- **Channel 1 (WE#):** 1A pin 2 from 74AHCT125 IC1 Gate 4 (Pico WE#, 5 V); 1B pin 3 from Z80 WR# (pin 22); 1Y pin 4 to SRAM WE# (pin 29).
- **Channel 2 (OE#):** 2A pin 5 from 74AHCT125 IC2 Gate 1 (Pico OE#, 5 V); 2B pin 6 from Z80 RD# (pin 21); 2Y pin 7 to SRAM OE# (pin 24).
- **Channel 3 (CE#):** 3A pin 11 from 74AHCT125 IC2 Gate 2 (Pico CE#, 5 V, new signal; Section 4); 3B pin 10 from Z80 MREQ# (pin 19); 3Y pin 9 to SRAM CE# (pin 22).
- **Channel 4:** Unused. Tie 4A (pin 14) and 4B (pin 13) to GND and leave 4Y (pin 12) open.
- **Strobe and power:** G (pin 15) to GND; VCC (pin 16) to +5V; GND (pin 8) to ground.

During Phase 6 bring-up with the Z80 absent, hold RESET# LOW (its Phase-1 default) so the AND gate forces the Pico "A" side regardless of the floating, pulled-up BUSACK# input; do not rely on BUSACK# alone during that phase.

Texas Instruments SN74HCT157 Data Selector/Multiplexer Datasheet Texas Instruments SN74HCT08 Quadruple 2-Input AND Gate Datasheet

1.2 SRAM Control-Source Arbitration: 74HCT157 and 74HCT08



2. 16-Bit Address Expansion Interface: MCP23S17-E/SP

The MCP23S17 acts as the dedicated 16-bit register shifter interfacing the Pico 2's SPI bus with the shared 5V address bus. During DMA block injection, it drives the target SRAM locations. During an active I/O trap, the transceiver directions are reversed, allowing the MCP23S17 to monitor the address bus states driven by the frozen Z80 CPU.

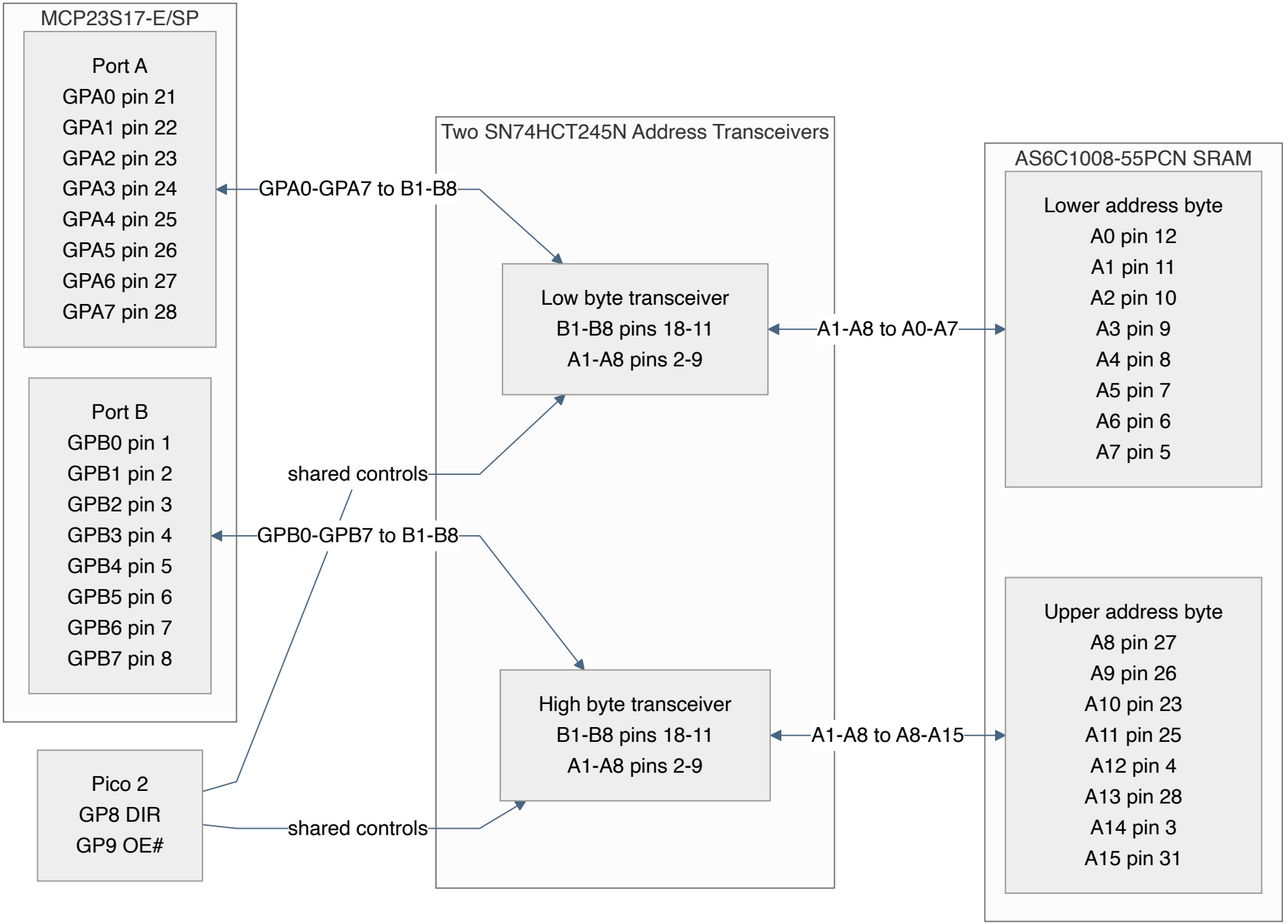
MCP23S17 Pin Designation	Target Connection	System Logic Role
GPA0 – GPA7 (Port A)	SN74HCT245N Transceiver #1 (B1 – B8)	Lower Address Byte Control ($A_0 – A_7$)
GPB0 – GPB7 (Port B)	SN74HCT245N Transceiver #2 (B1 – B8)	Upper Address Byte Control ($A_8 – A_{15}$)
CS# (Pin 11)	74AHCT125 IC3 Gate 1, from Pico GP21	SPI Hardware Chip Select (Active Low), 5V translated
CLK / SCK (Pin 12)	74AHCT125 IC3 Gate 2, from Pico GP18	SPI Master Clock Train Input, 5V translated
SI (Pin 13)	74AHCT125 IC3 Gate 3, from Pico GP19	SPI Master-Out-Slave-In (MOSI Path), 5V translated
SO (Pin 14)	SN74LVC244AN channel 2A1/2Y1 to Pico 2 GP20	SPI Master-In-Slave-Out (MISO Path), buffered from 5 V to 3.3 V; fit the Section 0.3 pull-up because SO is high-impedance while CS# is HIGH
A0, A1, A2 (Pins 15-17)	Tied to GND	Hardware address = 000; matches the fixed 0x40/0x41 opcode used in firmware regardless of the IOCON.HAEN state, and prevents floating address-select inputs
RESET# (Pin 18)	Tied to VCC (5V)	Hardware reset overridden for continuous software operation

Microchip MCP23017/MCP23S17 16-Bit I/O Expander with Serial Interface Datasheet

2.1 MCP23S17-to-SRAM Address Wiring

The MCP23S17 connects to the SRAM address inputs through two SN74HCT245N transceivers; it must not be connected directly to the shared address bus. Both transceivers share the Pico's GP8 direction control and GP9 active-low output-enable control described in Section 5.2. Side B faces the MCP23S17 and side A faces the SRAM and shared Z80 address bus.

2.1 MCP23S17-to-SRAM Address Wiring



3. Physical Partitioning & Breadboard Topology

The layout enforces a strict three-zone model across three 830-point breadboards to minimize cross-talk and propagation delay across the distinct 3.3V and 5V power domains. Each zone below lists the specific chips to place on that board and where to seat them.

- **Memory Board (Left Zone):** AS6C1008-55PCN SRAM, 74HCT157 SRAM control mux, 74HCT08 arbitration gate, and 74AHCT125N IC2. Put IC2 beside the mux because it supplies the Pico-side SRAM OE#/CE# inputs. Row budget: 16 (SRAM) + 8 (74HCT157) + 7 (74HCT08) + 7 (IC2) = 38 of 63 terminal rows.
- **Core Board (Center Zone):** Z84C0020PEC CPU, both SN74HCT245N address transceivers, 74AHCT125N IC1, and the SN74LVC245AN data transceiver. Keep IC1 beside the Z80 so its Gate 3 CLK output never crosses a board boundary. Put the address transceivers at the edge facing the Peripheral Board to shorten the MCP23S17's 16-bit feed, while their bus-facing side joins the local Z80 address bus and the adjacent SRAM. Row budget: 20 (Z80) + 20 (2x SN74HCT245N) + 7 (IC1)
    - 10 (SN74LVC245AN) = 57 of 63 terminal rows. *No hardware wait-state latches or flip-flops are used.*
- **Peripheral Board (Right Zone):** Raspberry Pi Pico 2, MCP23S17, 74AHCT125N IC3, and the SN74LVC244AN input buffer. Put the MCP23S17 and IC3 together and the LVC244 nearest the Core Board for the four Z80 status/control inputs; its fifth input, MCP SO, remains local. Row budget: 20 (Pico 2) + 14 (MCP23S17) + 7 (IC3) + 10 (SN74LVC244AN) = 51 of 63 terminal rows.

Use one supply-entry point and fan out +5 V and GND to each board; do not daisy-chain the boards' power rails end-to-end. Run the Pico 3.3 V rail separately to the two LVC devices. Bond adjacent boards with multiple short ground jumpers, especially beside the address/data bus crossings and CLK. Verify every BB830 distribution rail end-to-end with a meter before fitting links; never assume visually aligned rail segments are internally continuous.

3.1 High-Speed Interconnect Routing

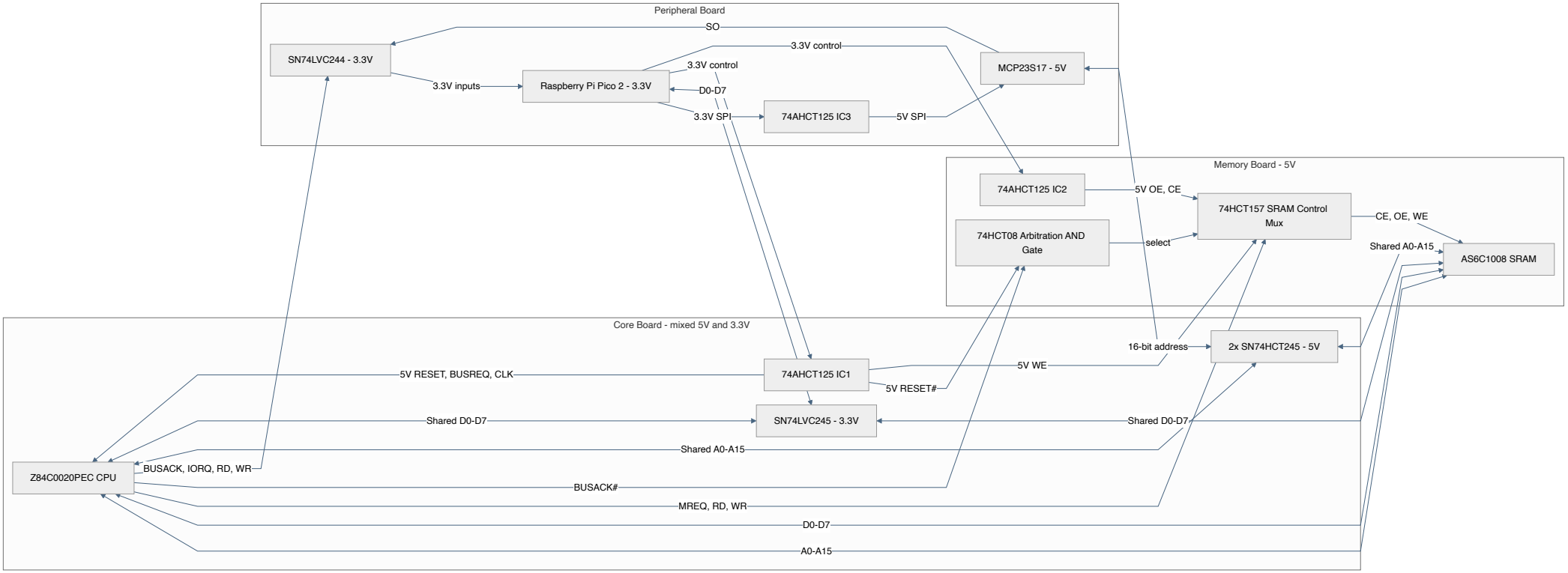
At the qualified 1-4 MHz clock rates, propagation skew from a few millimetres of wire-length difference is negligible compared with the Z80 timing budget. Solderless-breadboard reliability is instead dominated by total wire length, stubs, loop area, contact resistance, and fast-edge ringing. Route each bus as a short grouped trunk with roughly similar paths, but do not add serpentine wire merely to make lengths equal:

- **A0-A15:** keep the Z80, SRAM, and SN74HCT245N taps on one short trunk; avoid star branches and long unterminated stubs.
- **D0-D7:** group the Z80, SRAM, and SN74LVC245AN runs and keep the transceiver tap short.
- **74HCT157 outputs to SRAM CE#/OE#/WE#:** keep all three local to the SRAM and mux; exact length matching is unnecessary.
- **CLK (74AHCT125 IC1 Gate 3 output to Z80 pin 6):** route this as the single shortest, most direct jumper, kept away from the address bus. Do not lengthen it to match other nets; clock is the most edge-rate-sensitive signal in the design.

Route a ground jumper alongside every inter-board signal group and add ground probe points near CLK, IORQ#, MREQ#, RD#, WR#, SRAM CE#/OE#/WE#, and each bus transceiver. Keep all jumpers as short as the placement allows.

3.2 Major Chip Interconnection Overview

3.2 Major Chip Interconnection Overview



4. Level-Shifter Gate Mapping: 74AHCT125N (DIP-14)

Unidirectional signals generated by the 3.3 V Pico 2 are stepped up by three 5 V-powered 74AHCT125N arrays. Their TTL-compatible inputs accept a 3.3 V HIGH, while their lightly loaded outputs are guaranteed at least 4.4 V HIGH at VCC = 4.5 V, satisfying the Z80 clock input's stricter CMOS threshold. Nine Pico-driven signals need translation (RESET#, BUSREQ#, CLK, SRAM WE#/OE#/CE#, and SPI CS#/SCK/SI), which is one more than the two original packages could hold; IC3 was added to carry the remaining SRAM and SPI gates (Section 0.1).

IC Designation	Gate No.	Input Pin (3.3V from Pico)	Output Pin (5V to Target)	Functional Block Allocation
IC1: Buffer Array A	Gate 1	Pin 2 (from Pico GP3)	Pin 3 (to Z80 Pin 26 RESET#)	System Reset Generation
	Gate 2	Pin 5 (from Pico GP4)	Pin 6 (to Z80 Pin 25 BUSREQ#)	DMA Request Line
	Gate 3	Pin 9 (from Pico GP2)	Pin 8 (to Z80 Pin 6 CLK)	Master Clock Pulse Train
	Gate 4	Pin 12 (from Pico GP22)	Pin 11 (to 74HCT157 pin 2, channel 1 "A"; Section 1.2)	SRAM Write Enable (DMA side, via mux)
IC2: Buffer Array B	Gate 1	Pin 2 (from Pico GP26)	Pin 3 (to 74HCT157 pin 5, channel 2 "A"; Section 1.2)	SRAM Output Enable (DMA side, via mux)
	Gate 2	Pin 5 (from Pico GP5)	Pin 6 (to 74HCT157 pin 11, channel 3 "A"; Section 1.2)	SRAM Chip Enable (DMA side, via mux; new signal)
	Gates 3-4	Tied to GND	Leave Open	Unused channels grounded to prevent oscillation
IC3: Buffer Array C	Gate 1	Pin 2 (from Pico GP21)	Pin 3 (to MCP23S17 Pin 11 CS#)	SPI Chip Select Translation
	Gate 2	Pin 5 (from Pico GP18)	Pin 6 (to MCP23S17 Pin 12 SCK)	SPI Clock Translation
	Gate 3	Pin 9 (from Pico GP19)	Pin 8 (to MCP23S17 Pin 13 SI)	SPI MOSI Translation
	Gate 4	Tied to GND	Leave Open	Unused channel grounded to prevent oscillation

Tie every OE# input (pins 1, 4, 10, and 13 on each IC) to GND. For an unused gate, also tie its data input to GND and leave its output open, as shown above; the grounded OE# input then enables only a harmless LOW on that unconnected output.

The Pico-side default resistors in Section 0.3 are mandatory because the used AHCT125 gates are permanently enabled. Pulling only their 5 V outputs HIGH does not prevent the RP2350 reset-state pull-downs from driving BUSREQ#, SRAM CE#/OE#/WE#, and SPI CS# active before firmware starts.

5. Transceiver Operating Modes & Isolation Tables

The transceivers isolate the supervisor elements (Pico 2 and MCP23S17) from the main bus during standard execution, preventing bus contention.

5.1 SN74LVC245AN Data Bus Transceiver (3.3V Powered)

The bus-facing outputs reach approximately 3.3 V HIGH. This is valid for the Z80's ordinary data inputs ($V_{IH} \ge 2.2\text{ V}$) and the AS6C1008 inputs; only the Z80 CLK pin uses the stricter $V_{IHC}=V_{CC}-0.6\text{ V}$ threshold and is therefore translated through 74AHCT125 IC1.

Wire side A to the shared 5 V data bus and side B to the Pico. This orientation is required by the DIR values used below and in Section 8.13; preserve bit order exactly:

Bus bit	LVC245 side A	LVC245 side B	Pico GPIO
D0	A1 pin 2	B1 pin 18	GP10
D1	A2 pin 3	B2 pin 17	GP11
D2	A3 pin 4	B3 pin 16	GP12
D3	A4 pin 5	B4 pin 15	GP13
D4	A5 pin 6	B5 pin 14	GP14
D5	A6 pin 7	B6 pin 13	GP15
D6	A7 pin 8	B7 pin 12	GP16
D7	A8 pin 9	B8 pin 11	GP17

Connect DIR pin 1 to GP6, OE# pin 19 to GP7, VCC pin 20 to the Pico 3.3 V rail, and GND pin 10 to common ground.

System Operating State	OE# (GP7)	DIR (GP6)	Signal Direction	Functional Role
Boot / Write Mode	0 (Low)	0 (Low)	Side B → Side A (Pico → Bus)	Pico streams virtual ROM blocks into the SRAM.
Readback / Trap Mode	0 (Low)	1 (High)	Side A → Side B (Bus → Pico)	Pico reads data bus during verification or OUT traps.
Active Execution (Run)	1 (High)	X (Don't Care)	High-Impedance (High-Z)	Z80 and SRAM handle data operations directly; Pico data bus isolated.

5.2 SN74HCT245N Address Bus Transceivers (5V Powered)

System Operating State	OE# (GP9)	DIR (GP8)	Signal Direction	Functional Role
DMA Injection Mode	0 (Low)	0 (Low)	Side B → Side A (Expander → Bus)	MCP23S17 dictates memory-injected target address variables.
Trap Address Read Mode	0 (Low)	1 (High)	Side A → Side B (Bus → Expander)	Reverses transceivers so MCP23S17 can read the active port.
Active Execution (Run)	1 (High)	X (Don't Care)	High-Impedance (High-Z)	Z80 drives system address lines directly; expander isolated.

5.3 SN74LVC244AN 5 V-to-3.3 V Input Buffer

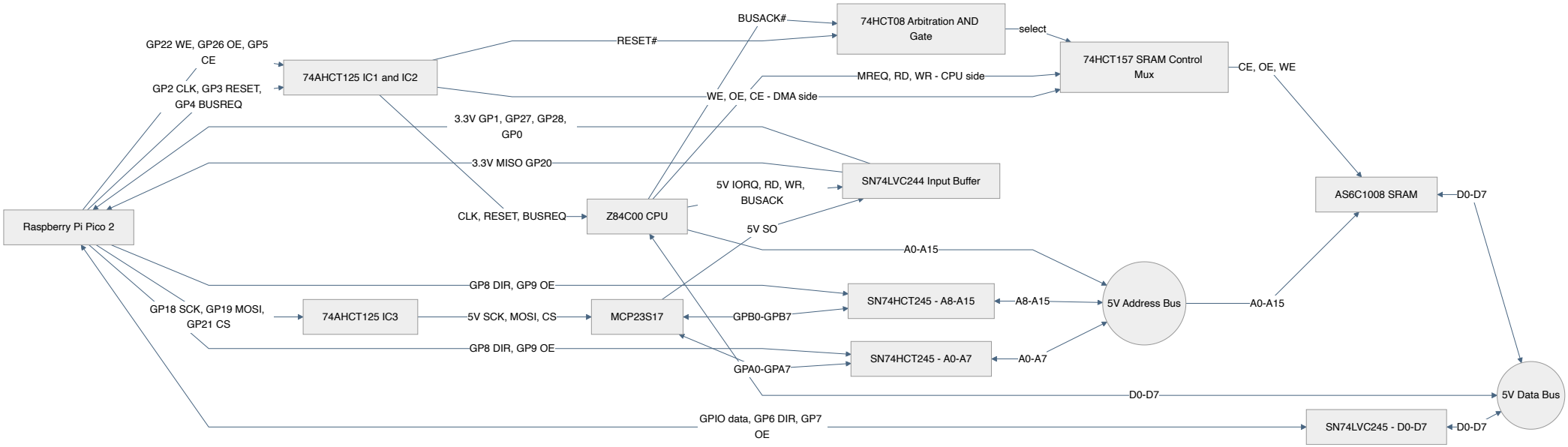
The RP2350's GP0-GP25 are 5 V-tolerant FT pads, but the 5.5 V rating applies only while IOVDD is powered at 3.3 V; with IOVDD at 0 V their absolute maximum is 3.63 V. GP26-GP29 are the QFN-60 package's ADC-capable pads and are not FT at all. Buffering all five incoming 5 V signals therefore avoids a power-sequencing constraint and keeps every Pico GPIO within its normal 3.3 V domain. Power the SN74LVC244AN from the Pico 3.3 V rail. Its inputs accept up to 5.5 V and its `Ioff` specification protects both sides when VCC is 0 V. Tie both output enables (pins 1 and 19) to GND: every buffered signal -- BUSACK#, IORQ#, RD#, WR#, and MCP SO -- must always be readable, and none of them share a Pico GPIO with any other driver, so neither output enable needs gating. Wire the channels as follows; tie every unused input to GND and leave unused outputs open.

5 V source	LVC244 input	3.3 V output	Pico destination
Z80 BUSACK# pin 23	1A1 pin 2	1Y1 pin 18	GP0
Z80 IORQ# pin 20	1A2 pin 4	1Y2 pin 16	GP1
Z80 RD# pin 21	1A3 pin 6	1Y3 pin 14	GP27
Z80 WR# pin 22	1A4 pin 8	1Y4 pin 12	GP28
MCP23S17 SO pin 14	2A1 pin 11	2Y1 pin 9	GP20
GND	2A2/2A3/2A4 pins 13/15/17	2Y2/2Y3/2Y4 pins 7/5/3 open	Unused

VCC (pin 20) connects to 3.3 V and GND (pin 10) to common ground. The 5 V-side pull-ups in Section 0.3 keep every input defined when its source is absent or high-impedance.

5.4 Bus and Supervisor Signal Interconnections

5.4 Bus and Supervisor Signal Interconnections



6. Architectural Operational Boundaries

6.1 Synchronous Clock-Stop Trap Protocol

When the Z80 executes an I/O instruction, the falling edge of IORQ# trips a hardware edge interrupt on the Pico 2. The handler disables the hardware PWM clock slice after GPIO synchronization and interrupt-entry latency. Because the Z84C00 is fully static, the clock can then remain stopped indefinitely in either a HIGH or LOW state. The design has no hardware WAIT# or combinational clock gate, so safe trap frequency is a measured property, not an assumption.

6.2 Terminal I/O over Pico WebSocket

The final terminal is a virtual I/O peripheral implemented by the Pico, not a Z80-side UART. Z80 software performs ordinary `IN` and `OUT` instructions against a small terminal port pair; the Pico intercepts those cycles through the Section 6.1 clock-stop trap, then resumes the CPU after sampling or supplying the data byte. A browser connects to the Pico over Wi-Fi and receives the terminal stream through a WebSocket server running on the Pico 2 W.

The WebSocket server must run on the Pico's other core so Wi-Fi, lwIP, HTTP serving, and WebSocket polling cannot interfere with the timing of the Z80-facing supervisor path. Core 0 owns GPIO, MCP23S17 SPI, clock stop/resume, DMA, and the I/O trap. Core 1 owns Wi-Fi association, the embedded terminal page, WebSocket accept/send/receive, and network polling. The cores share terminal byte queues, one immutable disk-write queue, a small request/result pair for Z80 bus ownership, and atomic terminal/disk status words; they share no raw bus GPIO ownership.

Use the `pico-altair-8800` WebSocket console as the software pattern: one embedded HTML terminal page, one WebSocket server, and byte queues between the CPU-facing side and the network-facing side. The trap hook must never call lwIP, print, sleep, wait for a browser, or block on a queue. It may only enqueue Z80 output bytes, dequeue already-buffered browser input bytes, and report terminal status. Core 1 may drop, drain, or back-pressure network data; it must not take locks needed by the trap or touch any Z80 bus GPIO.

The initial terminal decode is intentionally small:

Port	Direction	Function
`0x00`	Z80 `OUT`	Terminal transmit byte from Z80 to browser
`0x00`	Z80 `IN`	Terminal receive byte from browser to Z80; returns `0x00` if empty
`0x01`	Z80 `IN`	Terminal status: bit 0 = receive byte available, bit 1 = transmit queue has room, bit 7 = WebSocket client connected

This is enough for ROM monitors, BASIC, CP/M console glue, and small diagnostics. If later software needs modem-control style signals, add more virtual status bits before adding hardware.

6.3 Onboard Flash CP/M Disk Storage

Boot images and CP/M disk images live in a reserved region of the Pico's own onboard flash, not on any external card or bus. This needs zero extra GPIO and zero extra parts: the same physical flash chip that already holds the Pico's own firmware is memory-mapped and directly readable by both cores at any time, so no SPI bus, socket, or level-shifted chip-select is involved. GP27 and GP28 keep their original job monitoring Z80 RD# and WR# (Section 5.3); nothing needs reclaiming for storage.

Capacity budget. The Pico 2 W's onboard flash is 4 MiB. Reserve the top 1.4375 MiB for a power-fail journal, a distinct Z80 boot package, and four complete 320 KiB disks, leaving 2.5625 MiB for Pico firmware. That still comfortably covers this project's code plus the Wi-Fi firmware/CLM blob embedded by `pico_cyw43_arch`.

Region	Flash offset (from flash base)	Size	Contents
Journal	`0x290000`	64 KiB	Eight rotating metadata/data sector pairs
Boot	`0x2A0000`	128 KiB	Manifest plus a maximum 64 KiB Z80 RAM image

Region	Flash offset (from flash base)	Size	Contents
Drive A	0x2C0000	320 KiB	CP/M system disk
Drive B	0x310000	320 KiB	CP/M data disk
Drive C	0x360000	320 KiB	CP/M data disk
Drive D	0x3B0000	320 KiB	CP/M data disk

Each 320 KiB slot is exactly eighty 4 KiB flash erase sectors with no partial sector left over, so every slot boundary is also a sector boundary.

Disk-image contract. Here, "320 KiB" means exactly 327,680 bytes: 2,560 linear 128-byte CP/M records. The default BIOS presents those as 80 logical tracks of 32 records, numbered 1-32, with no skew. This is a project-specific logical geometry, not the IBM 3740 8-inch SSSD format (77 tracks x 26 records x 128 bytes = 256,256 bytes); existing images in another 8-inch format must be converted rather than copied into a slot unchanged. The matching CP/M 2.2 DPB is `SPT=32`, `BSH=4`, `BLM=15`, `EXM=1`, `DSM=155`, `DRM=63`, `AL0=0x80`, `AL1=0x00`, `CKS=0`, and `OFF=2`. Keep these values in the Z80 BIOS and host image builder from one shared generated definition.

Reservation mechanism. Set the CMake variable `PICO_FLASH_SIZE_BYTES` to `0x290000` before `pico_sdk_init()`. In Pico SDK 2.2 this value sizes the generated linker's `FLASH` region, while the `PICO_FLASH_SIZE_BYTES` C macro still comes from `pico2_w.h` and must remain the physical 4 MiB. That deliberate same-name split lets the linker reject code or `const` data that grows into storage while `hardware_flash` still accepts physical offsets through `0x3FFFFF`. Do not pass `-DPICO_FLASH_SIZE_BYTES=0x290000` to the compiler: its flash erase/program bounds would then reject every storage write. Add `static_assert(PICO_FLASH_SIZE_BYTES == 4u * 1024u * 1024u, "Pico 2 W flash size changed")` to the storage module and inspect the link map in CI to require `__flash_binary_end <= 0x10290000`.

The build must also link `pico_flash`, `hardware_flash`, `hardware_watchdog`, `pico_multicore`, and `pico_util`. Define `PICO_FLASH_ASSUME_CORE1_SAFE=1`: core 0 uses `flash_safe_execute()` only for journal recovery before core 1 is launched, while later core-1 writes still lock out core 0 normally. Core 0 must never erase/program flash after core 1 starts. The essential CMake ordering is:

```
set(PICO_FLASH_SIZE_BYTES 0x290000)
pico_sdk_init()

target_compile_definitions(z80_supervisor PRIVATE
  PICO_FLASH_ASSUME_CORE1_SAFE=1)
target_link_libraries(z80_supervisor PRIVATE
  pico_flash hardware_flash hardware_watchdog pico_multicore pico_util)
```

Provisioning. Build each disk as a flat, exactly-320-KiB binary and build the boot package described below, then write them outside the running firmware. Use `-v` so `picotool` verifies every load:

```
picotool load -v -o 0x102A0000 -t bin z80boot.pkg
picotool load -v -o 0x102C0000 -t bin drive-a.img
picotool load -v -o 0x10310000 -t bin drive-b.img
picotool load -v -o 0x10360000 -t bin drive-c.img
picotool load -v -o 0x103B0000 -t bin drive-d.img
```

(`0x10000000` is the Pico's XIP flash base address, so these absolute addresses are the flash-offset column above plus that base.) Reject any disk file whose host-side size is not exactly 327,680 bytes. If an RP2350 partition table is later embedded, declare these as data partitions and load them by partition ID; do not casually add `--ignore-partitions`. Firmware-only UF2 updates normally leave these addresses untouched, but `flash_nuke.uf2`, mass erase, or a replacement partition table destroys them, so keep host backups. There is no runtime bulk-image upload path in this design.

`z80boot.pkg` is little-endian. Its first 20 bytes are, in order, 32-bit magic `0x5442385A` ("Z8BT"), 16-bit version 1, 16-bit header length 20, 32-bit image length, 32-bit IEEE CRC32 of the image, and 32-bit IEEE CRC32 of the preceding 16 header bytes. Pad the remainder of the first 4 KiB with `0xFF`; the image begins at package offset `0x1000` and must be 1-65,536 bytes. The Z80 reset vector is at image offset zero. Generate this package and the BIOS DPB constants from one host tool so geometry and integrity metadata cannot drift.

Read path. A disk read is a plain pointer dereference into the memory-mapped flash region -- `XIP_BASE + drive_offset + sector_offset` -- with no SPI transaction, no DMA request, and no cross-core handshake. This is fast and safe to call directly from `io_trap_handler()` itself, unlike the deferred-queue pattern SD-based storage would have needed, provided no flash write (below) is concurrently in progress; the write path guarantees that by freezing the Z80 for its own duration, so a read trap can never race a write.

Write and recovery path. Updating one 128-byte CP/M record requires read-modify-erasing and reprogramming its enclosing 4 KiB flash block. The trap copies the complete 128-byte request into `disk_write_queue` and reports BUSY; it never erases flash itself. Core 1 constructs the new 4 KiB block in SRAM, asks core 0 to acquire BUSACK# while trapping remains enabled, and core 0 disables the trap only after BUSACK# is LOW. Core 1 then uses a bounded `flash_safe_execute()` call, which parks the registered core-0 victim and disables core-1 interrupts.

Each update rotates through one of eight 8 KiB journal pairs:

1. Erase the pair, then program the replacement 4 KiB data block.
2. Program a one-page header containing sequence, target offset, and CRC32. Until this valid header exists, the target remains untouched.
3. Erase/program the target block and verify all 4 KiB through XIP.
4. Erase the journal header only after verification succeeds.

At cold boot, core 0 scans valid journal headers before loading the Z80 image and restores them in sequence order. Therefore power loss before step 2 leaves the old target intact; power loss during or after step 3 leaves a valid replacement block from which boot recovery can finish. After a write, core 0 arms the trap while BUSACK# is still LOW and only then releases BUSREQ#, eliminating any interval in which the Z80 can run without I/O trapping. A bus-acquisition failure leaves the trap enabled; a release failure asserts RESET# and disables the trap. IORQ#, RD#, and WR# all tri-state whenever BUSACK# is asserted (Z8400 datasheet), so `io_trap_handler()` itself first confirms BUSACK# is still HIGH before touching anything; a stray edge while the Pico holds the bus disables the IORQ# interrupt and is ignored. The fitted 5 V-side 10 kOhm pull-ups in Section 0.3 keep the LVC244 inputs defined throughout the grant; Pico-side pulls cannot bias an input on the other side of the buffer. A successful write or recovery requires both a `PICO_OK` safe-execute result and callback verification of the flash contents. A runtime safe-execute failure asserts RESET#, isolates the buses, stops CLK, and forces a watchdog reboot before attempting another core-0 request: SDK lockout-exit failure can leave core 0 parked, so continuing to its release queue would deadlock instead of recovering.

Flash write endurance is finite and chip-specific; confirm the Pico 2 W's actual onboard flash part's rated erase-cycle endurance before relying on this for a write-heavy workload. This partition suits occasional CP/M saves and file writes well; it is a poor fit for a disk used as constant scratch/swap space.

Core assignment. Reads happen inline in `io_trap_handler()` on core 0, since they need no cross-core coordination. Writes are deferred: the trap enqueues a request, and core 1 -- the same task that owns the WebSocket terminal (Section 6.2) -- checks that queue on every loop iteration before doing network work, asks core 0's foreground loop to freeze the Z80, performs the journaled update, then releases it and reports READY or ERROR. Wi-Fi association is a bounded polling state machine, so storage remains available with no access point. Flash access never touches SPI0, the MCP23S17, or any Z80 bus GPIO.

6.4 System Performance Envelope & Constraints

- **I/O Decode Width:** Strictly limited to **8-bit** decoding (monitoring address lines A0-A7 via the lower expander port).
- **Trap Latency Profile:** Only the interval from IORQ# falling to the final PWM edge determines whether the Z80 is frozen in time; the subsequent SPI servicing time is unrestricted while the static CPU is stopped. Measure that initial interval at every claimed clock rate. The design is suitable for low-rate virtual peripherals, not high-speed line tracing.
- **Clock Validation Targets:**
    - *Bring-Up Target:* **1 MHz**, qualified only after a logic-analyzer capture proves every I/O cycle stops before the Z80 samples or releases its data.
    - *Qualification Range:* **2 MHz – 4 MHz**, tested in the increments in Section 8.12. No rate in this range is guaranteed in advance.

- *Failure Boundary:* If the PWM cannot be stopped in time at a desired rate, add hardware WAIT#/clock gating rather than relying on faster firmware. Operation above 4 MHz is outside this design's qualification plan.

7. Reference Firmware Implementations

This section previously duplicated firmware code with its own, ungrounded pin names, separate from the pin map and build phases in Section 8. To keep one definitive source, that code has been merged into Section 8.13, updated to the current pin numbers and to use 8.13's contention-safe helpers: the variable-frequency clock generator now appears under "Variable-Frequency Clock Generation" (Phase 2), bus acquisition under "Z80 Single-Step and Timed Bus Request" (Phases 7-8), the I/O trap ISR under "Synchronous I/O Trap Handler" (Phase 8), and the flash-backed image loader under "Flash Disk Image Loader" (Phase 9), and the queue-backed terminal bridge under "WebSocket Terminal I/O Bridge" (final Phase 10 integration).

The maintained, buildable firmware is now the canonical implementation: browse the source tree or use the complete source index. Section 8.13 remains a design-level reference for the safety invariants and integration order, but its excerpts should not be copied in place of the maintained source.

8. Progressive Build and Bring-Up Plan

Build and test one functional block at a time. Do not install the next chip until the current phase passes. Use sockets for all DIP devices, place a 100 nF ceramic capacitor directly across each IC's supply pins, and fit at least one 10 uF bulk capacitor per breadboard. Use a current-limited 5 V supply, multimeter, oscilloscope, and preferably a logic analyzer. Start each first power-up at a 100 mA current limit and remove power immediately if a rail falls by more than 5%, current rises unexpectedly, or a device becomes warm.

Fit the pull-ups and pull-downs in Section 0.3 so every signal has its specified state before firmware starts. Use temporary 1 kOhm series resistors when first connecting two potentially driven nodes. Record idle current after every phase. Unless stated otherwise, keep all chips from later phases out of their sockets.

8.1 Phase 0 - Empty Sockets and Power Distribution

Install: Breadboards, sockets, decoupling capacitors, pull-ups, power wiring, and signal wiring. Install no active device, including the Pico 2.

Implementation: Phase 0 power checklist.

Test plan:

1. With power disconnected, check resistance from each supply rail to ground. Investigate readings below 1 kOhm after capacitors charge.
2. Check every address, data, and control net end-to-end, then verify no continuity between neighboring bus lines.
3. Apply 5 V and measure every 5 V-powered DIP socket supply pin. Require 4.75 V to 5.25 V at VCC and less than 50 mV at each ground pin. The SN74LVC245AN and SN74LVC244AN VCC pins must remain at 0 V because their 3.3 V source, the absent Pico, is not yet installed.
4. Verify the external +5 V rail reaches Pico VSYS only through the 1N5819 and does not reach Pico VBUS, the 3.3 V rail, or any GPIO contact. With external power applied, VSYS must be one Schottky drop below the +5 V rail. Verify each 5 V-side active-low control is pulled HIGH. With power removed, measure approximately 10 kOhm from every Pico-side pull-up contact to the unpowered 3.3 V rail and from every pull-down contact to GND, as listed in Section 0.3; powered Pico-side logic levels are checked in Phase 1.

Pass gate: No shorts or crossed nets, correct supply voltage at every socket, and negligible current with all devices removed.

8.2 Phase 1 - Raspberry Pi Pico 2 Supervisor

Install: Pico 2 only.

Firmware feature: A diagnostic image must establish safe output levels before enabling any GPIO output: GP7 and GP9 HIGH to isolate the data and address transceivers; GP3 LOW to assert RESET#; GP4, GP5, GP21, GP22, and GP26 HIGH to deassert BUSREQ#, SRAM CE#, SPI CS#, SRAM WE#, and SRAM OE#; GP2 LOW to stop the clock; and GP6/GP8 LOW for the inactive transceiver directions. It must also provide a slow walking-one GPIO test selected through the USB serial console.

Implementation: Phase 1 application, backed by the shared supervisor module.

Test plan:

1. With the Section 0.4 1N5819 fitted, USB and external power may be connected together. Confirm neither source back-powers the other, then require 3.20 V to 3.40 V on the Pico 3.3 V rail and at the VCC socket contacts for the still-absent SN74LVC245AN and SN74LVC244AN.
2. Scope GP7 and GP9 through reset and startup; both must remain HIGH.
3. Verify GP3 (the future RESET# buffer input) is LOW and all other Pico control pins have the inactive levels listed above before, during, and after startup. The 5 V RESET# destination cannot be asserted until IC1 is installed in Phase 2.

4. Run the walking-one test and probe each destination socket. Require one-to-one routing, 0 V/3.3 V levels, and no change on neighboring pins. Restore safe levels when the test ends or the USB link drops.

Pass gate: Stable 3.3 V, safe startup levels, and correct routing for every Pico signal.

8.3 Phase 2 - 74AHCT125N Buffers

Install: IC1 first. Install IC2 only after IC1 passes. Keep the Z80 and SRAM removed. Tie every gate OE# pin LOW and every unused data input LOW; never leave an AHCT input floating.

Firmware feature: Add commands to toggle one buffered control at 10 Hz and generate selectable 1 kHz, 100 kHz, and 1 MHz 50% duty-cycle clocks on GP2.

Implementation: Phase 2 application, using the shared clock module.

Test plan:

1. Toggle one Pico source at a time. At its buffer output require LOW below 0.3 V, HIGH at or above 4.4 V, correct polarity, and no activity on adjacent outputs.
2. Test IC1 gate 3 at each clock frequency. Require 45% to 55% duty cycle and clean transitions at Z80 socket pin 6.
3. Verify RESET# becomes and remains LOW while BUSREQ#, SRAM CE#/WE#/ OE#, and SPI CS# remain HIGH for at least 100 ms after startup.
4. Power-cycle ten times while monitoring these signals. Any unintended transition fails the phase.

Pass gate: All translated outputs have valid 5 V levels and correct polarity, the 1 MHz clock is clean, and startup creates no active-low glitch.

8.4 Phase 3 - MCP23S17 SPI Address Generator

Install: The 3.3 V-powered SN74LVC244AN first, then 74AHCT125 IC3, then the MCP23S17-E/SP. Keep both SN74HCT245N devices removed so the MCP ports cannot drive the shared address bus.

Electrical hold point: Fit level translation on all SPI inputs. The MCP23S17 datasheet specifies $V_{IH} \geq 0.8V_{DD}$ for CS#, SCK, and SI, which is 4.0 V with a 5 V supply; a Pico 3.3 V HIGH is therefore not compliant. Translate Pico CS#, SCK, and MOSI up with 74AHCT125 IC3 (Section 4). Buffer MCP SO/MISO down through SN74LVC244AN channel 2A1/2Y1 (Section 5.3); do not connect it directly to GP20. Tie the MCP23S17 A0/A1/A2 hardware address pins to GND (Section 2). Confirm this wiring against the schematic before proceeding.

Firmware feature: Add byte-level SPI register read/write, a write-then-read register test, and 16-bit walking-one/walking-zero tests that can configure both MCP ports as either inputs or outputs.

Implementation: Phase 3 application, using the shared MCP23S17 driver.

Test plan:

1. Before fitting the MCP, pull each used LVC244 input LOW through 1 kOhm, then release it HIGH through its fitted 10 kOhm pull-up. Verify the matching Pico input reads LOW and HIGH at 3.3 V levels and unused channels do not change.
2. With SPI disconnected, verify MCP VDD, VSS, RESET#, and hardware address levels at the package.
3. With compliant translation fitted, write and read back 0x55 and 0xAA in IODIRA, IODIRB, OLATA, and OLATB.
4. Configure outputs and probe walking-one and walking-zero patterns at the empty SN74HCT245N sockets.
5. Configure inputs, apply 0 V or 5 V through 10 kOhm to each pin, and verify only the corresponding GPIO register bit changes.
6. Run 10,000 alternating register writes and reads with zero errors.

Pass gate: Compliant SPI levels, error-free register access, and correct operation of all 16 port bits in both directions.

8.5 Phase 4 - SN74HCT245N Address Transceivers

Install: The low-byte SN74HCT245N first, then the high-byte device after repeating and passing the tests. Keep the Z80 and SRAM removed.

Firmware feature: Add an address-bus test that always performs OE# HIGH, set DIR, set or sample the MCP ports, then OE# LOW. It must disable OE# before every direction change and on exit.

Implementation: Phase 4 application, using the shared bus module.

Test plan:

1. Before insertion, require HIGH at OE# pin 19. Verify the shared bus remains high-impedance after insertion.
2. Select MCP-to-bus direction and test 0x55, 0xAA, walking-one, and walking-zero patterns on every shared address line.
3. Disable OE#. Use a 10 kOhm test pull-up or pull-down to prove each bus line moves independently while the transceiver is disabled.
4. Select bus-to-MCP direction. Drive each bus line through 1 kOhm and verify the MCP reads it without driving back.
5. Repeat for the high byte, then test 0x0000, 0xFFFF, 0x5555, 0xAAAA, and a walking one across A0-A15.

Pass gate: Every address bit passes in both directions and both devices reliably enter high-impedance mode before DIR changes.

8.6 Phase 5 - SN74LVC245AN Data Transceiver

Install: SN74LVC245AN powered at 3.3 V. Keep the Z80 and SRAM removed. The TI datasheet permits inputs up to 5.5 V at this VCC.

Firmware feature: Add an 8-bit data-bus test using the same disable-change-enable sequence and fixed, walking-one, and walking-zero patterns on the Pico data GPIOs.

Implementation: Phase 5 application, using the shared bus module.

Test plan:

1. Require OE# HIGH before insertion and verify the bus-facing pins are high-impedance afterward.
2. Select Pico-to-bus direction and test 0x00, 0xFF, 0x55, 0xAA, walking-one, and walking-zero. Verify levels and bit order.
3. Disable OE#, reverse DIR, and re-enable. Drive each bus input with 0 V and 5 V through 1 kOhm and verify the Pico reading.
4. Run 1,000 disable-change-enable cycles while checking readback and supply current.

Pass gate: All eight bits pass both ways, isolation works, and no direction change causes contention or unexpected current.

8.7 Phase 6 - AS6C1008 SRAM and DMA Path

Install: The 74HCT08 arbitration gate first, then the 74HCT157 mux (Section 1.2), then the AS6C1008-55PCN. Keep the Z80 removed.

Electrical hold point: Confirm the 74HCT157 and 74HCT08 control-source arbitration from Section 1.2 is fitted. During DMA the Pico must exclusively control SRAM CE#, OE#, and WE# through the mux's "A" channels; during execution the Z80 must exclusively control them through MREQ#, RD#, and WR# on the mux's "B" channels. CE# must not be tied LOW in the finished system, and Pico and Z80 outputs must never be joined. With the Z80 removed for this phase, BUSACK# floats HIGH via its pull-up (Section 0.3); hold RESET# LOW (its Phase-1 default) so the 74HCT08 AND gate still forces the Pico "A" side regardless. Do not install SRAM until the mux and AND gate have passed static continuity and select-line tests.

Firmware feature: Add single-byte DMA read/write primitives, a walking address/data test, a two-pass full-memory pattern test, and a March C- or equivalent RAM test. Every failure must report its address, expected byte, and actual byte over USB serial.

Implementation: Phase 6 application, using the shared SRAM DMA module.

Test plan:

1. With transceivers disabled, require A16 LOW, CE2 HIGH, and CE#, OE#, and WE# HIGH directly at the SRAM.
2. In DMA mode, write and read one byte. Require WE# HIGH before address or data changes, and disable the Pico data driver before asserting OE# for readback.
3. Write unique values at 0x0000 and each power-of-two address from 0x0001 through 0x8000. Verify every value remains independent.
4. At several addresses test 0x00, 0xFF, 0x55, 0xAA, walking-one, and walking-zero data.
5. Fill all 65,536 bytes with the XOR of the address bytes, verify it, then repeat with the complement. Run the March test afterward.
6. Repeat after ten power cycles and with the intended 1 MHz timing.

Pass gate: Zero address, data, full-range pattern, or March-test errors and no overlap between CPU and Pico SRAM control sources.

8.8 Phase 7 - Z84C0020PEC CPU, Installed Last

Install: Z84C0020PEC. Preload SRAM first. Disable both bus transceivers, drive the Pico-side SRAM CE#/OE#/WE# controls inactive, hold RESET# LOW and BUSREQ# HIGH, and stop the clock LOW before insertion and power-up. RESET# LOW deliberately keeps the mux on the inactive Pico side; it switches to CPU controls automatically only when RESET# is released with BUSACK# HIGH.

Firmware feature: Add clock single-step and selectable 10 Hz, 1 kHz, 100 kHz, and 1 MHz run modes; reset pulse control; timed BUSREQ#/BUSACK# acquisition; and a command that preloads and verifies a small test program before releasing reset.

Implementation: Phase 7 application, using the shared CPU ownership module.

Test plan:

1. Verify RESET# LOW, BUSREQ# HIGH, both transceiver OE# signals HIGH, and SRAM write inactive immediately after power-up.
2. Clock at 10 Hz with RESET# asserted for at least three full cycles. Release reset and verify the first opcode fetch at 0x0000, including the expected M1#, MREQ#, and RD# sequence.
3. Preload `NOP; NOP; JP 0000h`. Single-step and verify address progression and the repeating jump while Pico bus drivers stay high-impedance.
4. Repeat at 1 kHz, 100 kHz, and 1 MHz with no unexpected SRAM writes.
5. Assert BUSREQ# while clocking. Require BUSACK# LOW after the current machine cycle and verify CPU bus outputs are high-impedance. Release BUSREQ# and require execution to resume.
6. Assert RESET# for at least three clocks during execution, release it, and verify a new fetch from 0x0000.

Pass gate: Correct reset fetch and execution through 1 MHz, valid BUSREQ#/BUSACK# transfer, and no bus contention.

8.9 Phase 8 - Integrated Virtual-ROM Boot and I/O Trap

Install: No further chips. Load final supervisor firmware.

Firmware feature: Combine safe startup, timed bus acquisition, image injection and readback, run control, and the synchronous IN/OUT trap. Maintain counters for boots, DMA failures, readback mismatches, trap timeouts, and unexpected RD#/WR# control states.

Implementation: Phase 8 application, using the shared I/O trap, CPU, and SRAM modules.

Test plan:

1. Acquire ownership by one of two complete procedures. Either use the timed BUSREQ#/BUSACK# handshake, or drive the Pico SRAM controls inactive, assert RESET#, continue the clock for at least three full cycles, and then stop it. In either case, verify the Z80 address and data buses are high-impedance before enabling a transceiver. The

Section 1.2 AND gate then grants the Pico exclusive SRAM control; inject a known 64 KB image, read back every byte, and compare it before CPU release.
2. Run a RAM increment loop for one hour at 1 MHz without corruption or supervisor timeout.
3. Execute OUT on ports 0x00, 0x01, 0x55, 0xAA, and 0xFF with matching data. Verify correct clock stop, address/data capture, driver disable, and clock resume.
4. Execute matching IN tests and verify each Pico-generated byte is stored correctly in SRAM.
5. With a logic analyzer, prove both transceiver OE# signals are HIGH during CPU cycles, the CPU bus is high-impedance before DMA enable, both DIR controls change only while their OE# is HIGH, and no overlap occurs between SRAM control sources.
6. Repeat cold boot, injection, execution, and I/O tests 100 times with zero logged failures.

Pass gate: Zero image or boot failures, correct IN/OUT behavior, one-hour stable execution, and contention-free ownership transitions.

8.10 Phase 9 - Flash Disk Image Loader and CP/M Storage

Install: No hardware rework. Set `PICO_FLASH_SIZE_BYTES` to `0x290000` in `CMakeLists.txt`, define `PICO_FLASH_ASSUME_CORE1_SAFE=1`, and link the libraries listed in Section 6.3. Provision the manifest-backed boot package and all four 320 KiB disk slots with the verified `picotool` commands there.

Firmware feature: With RESET# held LOW, recover any valid journal, validate the boot manifest and CRC32, DMA-write its payload to SRAM, and compare every byte before RESET# release. Do not wait for BUSACK# during this cold-boot path: RESET# itself selects the Pico's SRAM controls through Section 1.2's mux. Once running, ports `0x10-0x14` provide command/status, drive, 16-bit LBA, and 128-byte data transfers. Reads are synchronous XIP copies; writes use the journaled core-1 service and BUSY/READY/ERROR status defined in Section 8.13.

Implementation: Phase 9 application, with the shared disk device, flash backend, and flash layout.

Test plan:

1. With the Z80 socket populated and Phase 8 passing, confirm `io_trap_handler()` still passes every Phase 8 IN/OUT test unchanged; Phase 9 adds no new pins or rework to disturb it.
2. Confirm the link fails if code crosses `0x10290000`, while a runtime print/static assertion still reports the physical C macro as 4 MiB.
3. Provision the boot package and four exact-size disk images with `picotool -v`; read every region's first and last page back and compare host CRC32 values.
4. Cold-boot with RESET# held LOW. Require journal recovery and SRAM verification to finish without waiting for BUSACK#, then release RESET# only after success. Corrupt the manifest, payload, and SRAM readback separately and require each case to remain fail-closed.
5. Exercise all 2,560 LBAs on every drive through ports `0x10-0x14`. Verify exact 128-byte transfers, invalid drive/LBA rejection, command while BUSY rejection, and test-injected queue-full/error clearing behavior. After an injected flash or journal failure, require every READ/WRITE command to retain READY|ERROR until reboot recovery.
6. Probe BUSREQ#, BUSACK#, IORQ#, and CLK during a write. Require the trap to remain armed until BUSACK# is LOW and to be armed again before BUSACK# returns HIGH, with no untrapped I/O edge in either interval. Inject an IORQ# falling edge while BUSACK# is LOW and require the handler to disable the IRQ without touching CLK, SPI0, or either bus.
7. Add test-only power-cut hooks after journal-data program, header program, target erase, partial target program, target verification, and header clear. Reboot after every hook and require recovery to produce either the complete old block (before valid header) or the complete new block (after valid header), never a mixture.
8. Repeat reads and writes with Wi-Fi absent, associating, connected, and reconnecting. Disk completion must not depend on network state, and WebSocket queue overflow must remain counted rather than block.
9. Rewrite hot directory blocks repeatedly while tracking journal-pair rotation and the flash part's rated erase endurance. Treat this as a smoke test, not proof of lifetime.
10. Inject safe-execute entry and exit failures. Require RESET# LOW, isolated buses, stopped CLK, and a watchdog reboot without waiting on the core-0 release queue; recovery must retain the verified old or new disk block.

Pass gate: Boot and all four disks match host CRC32 values, every fault-injection reboot recovers an intact old or new block, all bounds and manifest failures remain fail-closed, disk service works without Wi-Fi, the linker protects the storage boundary, and logic-analyzer captures show no untrapped Z80 cycle around a flash write.

8.11 Phase 10 - WebSocket Terminal Console

Install: No further bus hardware. Use a Pico 2 W for the final networked terminal build, or keep the same firmware hooks compiled as stubs on a non-W Pico 2.

Firmware feature: Start the WebSocket console service on core 1 after core 0 has completed safe GPIO startup, queue initialization, and the Phase 9 boot-image load (which finishes entirely on core 0 before core 1 is launched). Core 0 continues to own the Z80 clock, bus transceivers, MCP23S17, SRAM DMA, and I/O trap. Core 1 owns Wi-Fi connection management, the embedded HTTP terminal page, WebSocket client state, network polling, and the Section 6.3 flash disk-write service -- all in the same `core1_main()` task, since `multicore_launch_core1()` only accepts one entry point. The two cores exchange only terminal bytes, flash disk-write requests, and status through nonblocking queues, following the `pico-altair-8800` console bridge pattern.

Implementation: Phase 10 application, with the shared terminal bridge, network service, and browser terminal.

Test plan:

1. Boot with no browser connected. Verify the Z80 still runs the Phase 8 I/O tests, `IN 0x01` reports no client, and terminal output does not accumulate without bound.
2. Connect a browser to `http://<pico-ip>:8088/`, or a WebSocket client to `ws://<pico-ip>:8088/`. Verify `IN 0x01` sets the client-connected bit without disturbing the Z80 clock.
3. Run a Z80 program that writes a continuous alphabet pattern to `OUT 0x00`. Verify the browser receives the stream in order and that queue-full conditions are counted rather than blocking the trap.
4. Type from the browser and verify the Z80 receives each byte through `IN 0x00` only after `IN 0x01` reports data available.
5. Disconnect and reconnect the browser while the Z80 test program runs. Verify stale input is cleared, output resumes for the new client, and no trap timeout counter increments.
6. Exhaust the default alarm pool before service startup and require the supervisor to remain fail-closed rather than launching core 1 without both WebSocket polling timers.

Pass gate: The WebSocket service remains responsive while the Z80 runs at the Phase 8 qualified 1 MHz setting, no network path runs on the core that services Z80 timing, and all terminal queue overflow or client disconnect conditions are visible through counters rather than blocking the CPU trap.

8.12 Frequency Qualification

Begin only after Phase 10 passes at 1 MHz. Test 2 MHz, then increase in 500 kHz steps to 4 MHz. At each step repeat SRAM readback, the one-hour memory loop, and continuous IN/OUT tests while measuring stop latency. The qualified frequency is the highest error-free step for which the logic analyzer proves the clock always stops before the Z80 advances beyond the safe trap point. Do not claim operation above 4 MHz without equivalent timing evidence.

8.13 Pico Diagnostic Firmware Core Features

These Pico SDK fragments show the safety-critical core of each test and finish with the required two-core `main()` integration order. Board- specific Wi-Fi/WebSocket hooks and the optional nonblocking USB command parser remain external. Every diagnostic command must print `PASS` or a detailed failure and call `isolate_buses()` before returning. `PIN_BUSACK_N` is GP0, driven from Z80 BUSACK# (pin 23) through the SN74LVC244AN input buffer (Section 5.3); IORQ#, RD#, and MCP SO use the same buffer, so no 5 V output reaches the Pico directly. `PIN_SRAM_CE_N` is fixed at GP5 (74AHCT125 IC2 Gate 2; Section 4), moved off GP23, which on the Pico 2 W is dedicated to the CYW43439 wireless module's control interface (shared with GP24/GP25/GP29) and must never be repurposed. `PIN_DATA_0`- `PIN_DATA_7` occupy GP10-GP17. There is no `PIN_SRAM_SOURCE_SELECT` GPIO: the 74HCT157 select input is driven by a 74HCT08 AND gate combining RESET# and BUSACK# (Section 1.2), so it still switches automatically with no extra Pico pin. `PIN_WR_N` is GP28, buffered through the same SN74LVC244AN (Section 5.3); `io_trap_handler()` reads both `PIN_RD_N` and `PIN_WR_N` to resolve cycle intent.

Safe Startup and Walking Output (Phases 1-2)

Maintained source: pins.h, supervisor.h, and supervisor.c.

Preload each output latch while the pin is still an input, then enable the output driver. This prevents a brief LOW pulse on active-low lines.

```
#include <stdint.h>
#include <stdio.h>
#include "pico/stdlib.h"

enum {
  PIN_IORQ_N = 1, PIN_CLK = 2, PIN_RESET_N = 3,
  PIN_BUSREQ_N = 4, PIN_BUSACK_N = 0,
  PIN_DATA_DIR = 6, PIN_DATA_OE_N = 7,
  PIN_ADDR_DIR = 8, PIN_ADDR_OE_N = 9,
  PIN_DATA_0 = 10, PIN_DATA_1 = 11, PIN_DATA_2 = 12, PIN_DATA_3 = 13,
  PIN_DATA_4 = 14, PIN_DATA_5 = 15, PIN_DATA_6 = 16, PIN_DATA_7 = 17,
  PIN_SPI_SCK = 18, PIN_SPI_MOSI = 19, PIN_SPI_MISO = 20,
  PIN_SPI_CS_N = 21,
  PIN_SRAM_WE_N = 22, PIN_SRAM_CE_N = 5, PIN_SRAM_OE_N = 26,
  PIN_RD_N = 27, PIN_WR_N = 28
};

static void output_with_initial_level(uint pin, bool level) {
  gpio_init(pin);
  gpio_put(pin, level);       // Preload SIO output latch.
  gpio_set_dir(pin, GPIO_OUT);
}

static void input_with_no_pull(uint pin) {
  gpio_init(pin);
  gpio_set_dir(pin, GPIO_IN);
  gpio_disable_pulls(pin);
}

static void diagnostic_safe_startup(void) {
  output_with_initial_level(PIN_DATA_OE_N, 1);
  output_with_initial_level(PIN_ADDR_OE_N, 1);
  output_with_initial_level(PIN_RESET_N, 0); // Hold CPU reset.
  output_with_initial_level(PIN_BUSREQ_N, 1);
  output_with_initial_level(PIN_SRAM_WE_N, 1);
  output_with_initial_level(PIN_SRAM_CE_N, 1);
  output_with_initial_level(PIN_SRAM_OE_N, 1);
  output_with_initial_level(PIN_SPI_CS_N, 1);
  output_with_initial_level(PIN_CLK, 0);
  output_with_initial_level(PIN_DATA_DIR, 0);
  output_with_initial_level(PIN_ADDR_DIR, 0);
  input_with_no_pull(PIN_BUSACK_N);
  input_with_no_pull(PIN_IORQ_N); // Section 0.3 pulls up the LVC244 input.
  input_with_no_pull(PIN_RD_N);
  input_with_no_pull(PIN_WR_N);
}

static void walking_output_test(const uint *pins, size_t count) {
  for (size_t active = 0; active < count; ++active) {
    for (size_t index = 0; index < count; ++index)
      gpio_put(pins[index], index == active);
    sleep_ms(250);          // Probe or logic-analyzer capture point.
  }
  diagnostic_safe_startup();
}
```

Run `walking_output_test()` only in Phases 1-2 while the destination driver/bus chips are absent. It deliberately changes raw pin levels and is not safe as an in-system diagnostic after Phase 3.

Variable-Frequency Clock Generation (Phase 2, Phases 7-8 Run Modes)

Maintained source: clock.h and clock.c.

Configure `PIN_CLK` as a PWM output once during Phase 2 bring-up. Reuse the same slice for the selectable 10 Hz/1 kHz/100 kHz/1 MHz run modes in Phases 7-8, and to freeze the clock during the Phase 8 I/O trap.

```
#include "hardware/pwm.h"
#include "hardware/clocks.h"

static bool set_z80_clock_hz(uint32_t hz) {
  if (hz < 10 || hz > 4000000)
    return false;
```

```
  uint slice_num = pwm_gpio_to_slice_num(PIN_CLK);
  pwm_set_enabled(slice_num, false);
  uint32_t sys_clk = clock_get_hz(clk_sys);
  uint64_t sys_clk16 = (uint64_t)sys_clk * 16u;
  uint32_t best_divider16 = 0;
  uint32_t best_count = 0;
  uint64_t best_error = UINT64_MAX;

  // Search all legal 8.4 dividers; this runs only when frequency changes.
  for (uint32_t divider16 = 16; divider16 <= 4095; ++divider16) {
    uint64_t denominator = (uint64_t)hz * divider16;
    uint64_t count = (sys_clk16 + denominator / 2u) / denominator;
    if (count < 1) count = 1;
    if (count > 65536) count = 65536;
    uint64_t product = (uint64_t)divider16 * count;
    uint64_t target_product = (uint64_t)hz * product;
    uint64_t error = sys_clk16 > target_product ?
      sys_clk16 - target_product : target_product - sys_clk16;
    uint64_t best_product = (uint64_t)best_divider16 * best_count;
    if (best_divider16 == 0 ||
      error * best_product < best_error * product) {
      best_divider16 = divider16;
      best_count = (uint32_t)count;
      best_error = error;
    }
  }

  uint16_t wrap = (uint16_t)(best_count - 1u);
  pwm_set_clkdiv(slice_num, (float)best_divider16 / 16.0f);
  pwm_set_wrap(slice_num, wrap);
  pwm_set_chan_level(slice_num, PWM_CHAN_A,
    (uint16_t)(best_count / 2u)); // Exact 50% when count is even.
  pwm_set_counter(slice_num, 0);
  gpio_set_function(PIN_CLK, GPIO_FUNC_PWM);
  pwm_set_enabled(slice_num, true);
  return true;
}

static void stop_z80_clock(void) {
  pwm_set_enabled(pwm_gpio_to_slice_num(PIN_CLK), false); // May freeze HIGH or LOW.
}

static void resume_z80_clock(void) {
  pwm_set_enabled(pwm_gpio_to_slice_num(PIN_CLK), true);
}
```

MCP23S17 Register and Port Test (Phases 3-4)

Maintained source: mcp23s17.h and mcp23s17.c.

The SPI translator sits between these Pico pins and the 5 V MCP23S17; MISO/SO returns through the SN74LVC244AN. The register test proves communication before either address transceiver is enabled.

```
#include "hardware/spi.h"

enum { MCP_WRITE = 0x40, MCP_READ = 0x41 };
enum { IODIRA = 0x00, IODIRB = 0x01, GPIOA = 0x12, GPIOB = 0x13,
     OLATA = 0x14, OLATB = 0x15 };

static void mcp_spi_init(void) {
  output_with_initial_level(PIN_SPI_CS_N, 1);
  spi_init(spi0, 4000000);
  gpio_set_function(PIN_SPI_SCK, GPIO_FUNC_SPI);
  gpio_set_function(PIN_SPI_MOSI, GPIO_FUNC_SPI);
  gpio_set_function(PIN_SPI_MISO, GPIO_FUNC_SPI);
}

static void mcp_write(uint8_t reg, uint8_t value) {
  uint8_t frame[] = { MCP_WRITE, reg, value };
  gpio_put(PIN_SPI_CS_N, 0);
  spi_write_blocking(spi0, frame, sizeof frame);
  gpio_put(PIN_SPI_CS_N, 1);
}

static uint8_t mcp_read(uint8_t reg) {
  uint8_t tx[] = { MCP_READ, reg, 0x00 };
```

```
  uint8_t rx[sizeof tx];
  gpio_put(PIN_SPI_CS_N, 0);
  spi_write_read_blocking(spi0, tx, rx, sizeof tx);
  gpio_put(PIN_SPI_CS_N, 1);
  return rx[2];
}

static bool mcp_register_test(void) {
  const uint8_t patterns[] = { 0x55, 0xAA };
  mcp_write(IODIRA, 0x00);
  mcp_write(IODIRB, 0x00);
  for (size_t i = 0; i < sizeof patterns; ++i) {
    mcp_write(OLATA, patterns[i]);
    mcp_write(OLATB, (uint8_t)~patterns[i]);
    if (mcp_read(OLATA) != patterns[i] ||
      mcp_read(OLATB) != (uint8_t)~patterns[i])
      return false;
  }
  return true;
}
```

Contention-Safe Transceiver Changes (Phases 4-6)

Maintained source: bus.h and bus.c.

OE# must be HIGH before DIR changes. Keep this invariant in one helper instead of duplicating raw GPIO writes throughout the test firmware.

```
static void isolate_buses(void) {
  gpio_put(PIN_ADDR_OE_N, 1);
  gpio_put(PIN_DATA_OE_N, 1);
}

static void set_transceiver(uint oe_n, uint dir, bool direction) {
  gpio_put(oe_n, 1);
  busy_wait_us_32(1);
  gpio_put(dir, direction);
  busy_wait_us_32(1);
  gpio_put(oe_n, 0);
}
```

Data and Address Bus GPIO Helpers (Phases 4-6)

Maintained source: bus.h and bus.c.

`PIN_DATA_0`-`PIN_DATA_7` (GP10-GP17) carry D0-D7 through the SN74LVC245AN; direction and isolation reuse the existing `PIN_DATA_DIR`/`PIN_DATA_OE_N` pair and `set_transceiver()`/`isolate_buses()`. Address bytes reach the shared bus through the MCP23S17's own output latches, already exercised in the Phase 3-4 register test. Call these helpers only while RESET# is asserted or `request_cpu_bus()` has returned true, and keep SRAM CE# HIGH while changing address or data direction.

```
static const uint DATA_PINS[8] = {
  PIN_DATA_0, PIN_DATA_1, PIN_DATA_2, PIN_DATA_3,
  PIN_DATA_4, PIN_DATA_5, PIN_DATA_6, PIN_DATA_7
};

static void data_bus_drive(uint8_t value) {
  gpio_put(PIN_DATA_OE_N, 1);
  for (size_t i = 0; i < 8; ++i) {
    gpio_put(DATA_PINS[i], (value >> i) & 1);  // Preload before enabling output.
    gpio_set_dir(DATA_PINS[i], GPIO_OUT);
  }
  set_transceiver(PIN_DATA_OE_N, PIN_DATA_DIR, 0); // Pico -> Bus.
}

static void data_bus_prepare_input(void) {
  gpio_put(PIN_DATA_OE_N, 1);
  for (size_t i = 0; i < 8; ++i)
    gpio_set_dir(DATA_PINS[i], GPIO_IN);
  set_transceiver(PIN_DATA_OE_N, PIN_DATA_DIR, 1); // Bus -> Pico.
}

static uint8_t data_bus_sample(void) {
```

```
  uint8_t value = 0;
  for (size_t i = 0; i < 8; ++i)
    value |= gpio_get(DATA_PINS[i]) << i;
  return value;
}

static void address_bus_drive(uint16_t address) {
  gpio_put(PIN_ADDR_OE_N, 1);
  mcp_write(OLATA, (uint8_t)address);
  mcp_write(OLATB, (uint8_t)(address >> 8));
  mcp_write(IODIRA, 0x00);
  mcp_write(IODIRB, 0x00);
  set_transceiver(PIN_ADDR_OE_N, PIN_ADDR_DIR, 0); // Expander -> Bus.
}
```

SRAM DMA Access and Pattern Test (Phase 6)

Maintained source: sram.h and sram.c.

The address and data helper functions above represent the already tested MCP23S17 and GPIO bus operations. The control-source selector must grant exclusive SRAM control to the Pico before this code runs.

```
static void dma_write_byte(uint16_t address, uint8_t value) {
  gpio_put(PIN_SRAM_CE_N, 1);
  gpio_put(PIN_SRAM_OE_N, 1);
  gpio_put(PIN_SRAM_WE_N, 1);
  address_bus_drive(address);       // MCP -> HCT245 -> A0-A15.
  data_bus_drive(value);            // Pico -> LVC245 -> D0-D7.
  gpio_put(PIN_SRAM_CE_N, 0);
  busy_wait_us_32(1);
  gpio_put(PIN_SRAM_WE_N, 0);
  busy_wait_us_32(1);
  gpio_put(PIN_SRAM_WE_N, 1);
  gpio_put(PIN_SRAM_CE_N, 1);
}

static uint8_t dma_read_byte(uint16_t address) {
  uint8_t value;
  gpio_put(PIN_SRAM_CE_N, 1);
  gpio_put(PIN_SRAM_WE_N, 1);
  address_bus_drive(address);
  data_bus_prepare_input();         // Disable, reverse, then enable.
  gpio_put(PIN_SRAM_CE_N, 0);
  gpio_put(PIN_SRAM_OE_N, 0);
  busy_wait_us_32(1);
  value = data_bus_sample();
  gpio_put(PIN_SRAM_OE_N, 1);
  gpio_put(PIN_SRAM_CE_N, 1);
  gpio_put(PIN_DATA_OE_N, 1);
  return value;
}

static bool sram_pattern_test(bool complement) {
  for (uint32_t address = 0; address < 65536; ++address) {
    uint8_t expected = (uint8_t)address ^ (uint8_t)(address >> 8);
    dma_write_byte((uint16_t)address,
             complement ? (uint8_t)~expected : expected);
  }
  for (uint32_t address = 0; address < 65536; ++address) {
    uint8_t expected = (uint8_t)address ^ (uint8_t)(address >> 8);
    if (complement)
      expected = (uint8_t)~expected;
    uint8_t actual = dma_read_byte((uint16_t)address);
    if (actual != expected) {
      printf("FAIL %04lx expected=%02x actual=%02x\n",
           (unsigned long)address, expected, actual);
      isolate_buses();
      return false;
    }
  }
  isolate_buses();
  return true;
}
```

Z80 Single-Step and Timed Bus Request (Phases 7-8)

Maintained source: cpu.h and cpu.c.

Single-step uses SIO rather than PWM. `request_cpu_bus()` and `release_cpu_bus()` each take an explicit `timeout_us` bound so a wiring fault or a missing Z80 cannot hang the supervisor waiting on BUSACK#. The clock must be running while requesting or releasing BUSREQ#; after using `clock_one_cycle()`, call `set_z80_clock_hz()` to restore the CLK pin's PWM function before either handshake.

```
static void clock_one_cycle(uint32_t half_period_us) {
  pwm_set_enabled(pwm_gpio_to_slice_num(PIN_CLK), false);
  gpio_set_function(PIN_CLK, GPIO_FUNC_SIO);
  gpio_set_dir(PIN_CLK, GPIO_OUT);
  gpio_put(PIN_CLK, 0);
  busy_wait_us_32(half_period_us);
  gpio_put(PIN_CLK, 1);
  busy_wait_us_32(half_period_us);
  gpio_put(PIN_CLK, 0);
}

static bool request_cpu_bus(uint32_t timeout_us) {
  absolute_time_t deadline = make_timeout_time_us(timeout_us);
  isolate_buses();
  gpio_put(PIN_BUSREQ_N, 0);
  while (gpio_get(PIN_BUSACK_N) != 0) {
    if (time_reached(deadline)) {
      gpio_put(PIN_BUSREQ_N, 1);
      return false;
    }
    tight_loop_contents();
  }
  return true;               // RESET# HIGH: BUSACK# LOW selects DMA controls.
}

static bool release_cpu_bus(uint32_t timeout_us) {
  absolute_time_t deadline = make_timeout_time_us(timeout_us);
  gpio_put(PIN_SRAM_CE_N, 1);
  gpio_put(PIN_SRAM_OE_N, 1);
  gpio_put(PIN_SRAM_WE_N, 1);
  isolate_buses();
  gpio_put(PIN_BUSREQ_N, 1);
  while (gpio_get(PIN_BUSACK_N) == 0) {
    if (time_reached(deadline)) {
      gpio_put(PIN_RESET_N, 0);
      return false;
    }
    tight_loop_contents();
  }
  return true;               // BUSACK# HIGH restores CPU controls.
}
```

Synchronous I/O Trap Handler (Phase 8)

Maintained source: io_trap.h and io_trap.c.

A falling-edge IRQ on `PIN_IORQ_N` freezes the clock, reverses the address transceiver with the same contention-safe helper used elsewhere, reads the trapped port from the lower MCP23S17 port (Section 6.4's 8-bit decode limit), and reuses the already-tested data bus helpers from the SRAM DMA code to sample or drive the data byte. The handler first confirms `PIN_BUSACK_N` is still HIGH; IORQ# tri-states along with RD#/WR# whenever BUSACK# is asserted (Section 6.3), so a falling edge seen while the Pico already owns the bus cannot be a real Z80 cycle and is ignored before any bus or SPI0 state is touched. Both MCP ports must be forced to inputs before the two transceivers' shared OE# is enabled, even though only GPIOA is read; otherwise the still-output GPB port fights Z80 A8-A15. For `IN`, the data byte must stay driven until the Z80 samples it, so the clock resumes and `RD#` is polled before the data bus is isolated.

`process_virtual_io_read` and `process_virtual_io_write` are the only application-supplied hooks. They must use the signatures below, finish without sleeping, printing, waiting on USB, or taking a lock held by main code, and must not start another bus operation. Move expensive work to the main loop through a lock-free queue.

```
#include "hardware/watchdog.h"
```

```c
uint8_t process_virtual_io_read(uint8_t port);
void process_virtual_io_write(uint8_t port, uint8_t value);

enum { TRAP_RELEASE_TIMEOUT_US = 500000 }; // Covers the 10 Hz test mode.
static volatile uint32_t trap_timeout_count;
static volatile uint32_t unexpected_control_count;

static _Noreturn void reset_after_trap_fault(void) {
  isolate_buses();
  gpio_put(PIN_RESET_N, 0);
  stop_z80_clock();
  for (unsigned int cycle = 0; cycle < 3; ++cycle)
    clock_one_cycle(1); // RESET# setup and each half-cycle exceed Z80 minima.
  watchdog_reboot(0, 0, 0); // Fail-closed: same recovery path as flash faults.
  while (true)
    tight_loop_contents(); // watchdog_reboot() takes effect asynchronously.
}

static void io_trap_handler(uint gpio, uint32_t events) {
  (void)events;
  if (gpio != PIN_IORQ_N)
    return;  // Only IORQ# is ever armed, but never trust a shared callback.
  if (!gpio_get(PIN_BUSACK_N)) {
    gpio_set_irq_enabled(PIN_IORQ_N, GPIO_IRQ_EDGE_FALL, false);
    return;  // The foreground release path re-arms this interrupt.
  }
  stop_z80_clock();
  mcp_write(IODIRA, 0xFF);
  mcp_write(IODIRB, 0xFF);                          // Both share the enabled transceiver path.
  set_transceiver(PIN_ADDR_OE_N, PIN_ADDR_DIR, 1); // Bus -> Expander.
  uint8_t port = mcp_read(GPIOA);                   // 8-bit I/O decode.
  gpio_put(PIN_ADDR_OE_N, 1);                       // Address captured; isolate now.
  mcp_write(IODIRA, 0x00);                          // Restore DMA output mode.
  mcp_write(IODIRB, 0x00);

  bool is_read = !gpio_get(PIN_RD_N);
  bool is_write = !gpio_get(PIN_WR_N);
  if (is_read == is_write) {      // Neither or both asserted: hardware fault.
    ++unexpected_control_count;
    is_write = !is_read;          // Best-effort fallback so the trap still resolves.
  }

  if (is_write) {
    data_bus_prepare_input();                       // Bus -> Pico.
    uint8_t value = data_bus_sample();
    isolate_buses();
    process_virtual_io_write(port, value);
    resume_z80_clock();
  } else {
    data_bus_drive(process_virtual_io_read(port));     // Pico -> Bus.
    absolute_time_t deadline = make_timeout_time_us(TRAP_RELEASE_TIMEOUT_US);
    resume_z80_clock();
    while (!gpio_get(PIN_RD_N)) {                     // Hold data through the sample edge.
      if (time_reached(deadline)) {
        ++trap_timeout_count;
        reset_after_trap_fault();
        return;
      }
      tight_loop_contents();
    }
    isolate_buses();
  }
}

static void enable_io_trap(void) {
  input_with_no_pull(PIN_IORQ_N); // Pull-up is on the LVC244's 5 V input.
  gpio_acknowledge_irq(PIN_IORQ_N, GPIO_IRQ_EDGE_FALL);
  gpio_set_irq_enabled_with_callback(PIN_IORQ_N, GPIO_IRQ_EDGE_FALL,
    true, &io_trap_handler);
}

static void disable_io_trap(void) {
  gpio_set_irq_enabled(PIN_IORQ_N, GPIO_IRQ_EDGE_FALL, false);
  gpio_acknowledge_irq(PIN_IORQ_N, GPIO_IRQ_EDGE_FALL);
}
```

Flash Disk Image Loader (Final Phase 9 Integration)

Maintained source: flash_disk.h, flash_layout.h, disk_device.c, and flash_backend.c.

Loading a boot image is now a synchronous, core-0-only operation: a flash read is an ordinary memory access through the XIP-mapped pointer (Section 6.3), so no filesystem, blocking I/O call, or core 1 task is needed to bring the initial image into SRAM, unlike the SD design this replaces. Only CP/M's live disk-sector *writes* still need to run on core 1 and cross back to core 0's foreground loop, because only they need to freeze the Z80 around a flash erase/program cycle (Section 6.3).

```
#include <stdint.h>
#include <stdbool.h>
#include <stddef.h>
#include <stdio.h>
#include <string.h>

#include "hardware/flash.h"
#include "hardware/watchdog.h"
#include "pico/flash.h"
#include "pico/util/queue.h"

enum {
  FLASH_JOURNAL_BASE_OFFSET = 0x290000u,
  FLASH_JOURNAL_BYTES = 0x10000u,
  FLASH_JOURNAL_PAIR_BYTES = 2u * FLASH_SECTOR_SIZE,
  FLASH_JOURNAL_PAIR_COUNT = FLASH_JOURNAL_BYTES / FLASH_JOURNAL_PAIR_BYTES,
  FLASH_BOOT_BASE_OFFSET = 0x2A0000u,
  FLASH_BOOT_REGION_BYTES = 0x20000u,
  FLASH_BOOT_PAYLOAD_OFFSET = FLASH_BOOT_BASE_OFFSET + FLASH_SECTOR_SIZE,
  FLASH_DISK_BASE_OFFSET = 0x2C0000u,   // Section 6.3 partition table.
  FLASH_DISK_SLOT_BYTES = 0x50000u,     // 320 KiB per drive.
  FLASH_DISK_SLOT_COUNT = 4u,
  FLASH_DISK_RECORD_BYTES = 128u,
  FLASH_DISK_RECORD_COUNT = 2560u,
  SRAM_SIZE_BYTES = 65536
};

_Static_assert(PICO_FLASH_SIZE_BYTES == 4u * 1024u * 1024u,
  "Pico 2 W physical flash size changed");
_Static_assert(FLASH_DISK_SLOT_BYTES ==
  FLASH_DISK_RECORD_BYTES * FLASH_DISK_RECORD_COUNT,
  "disk geometry does not fill its slot");
_Static_assert(FLASH_BOOT_PAYLOAD_OFFSET + SRAM_SIZE_BYTES <=
  FLASH_BOOT_BASE_OFFSET + FLASH_BOOT_REGION_BYTES,
  "boot payload exceeds its reserved region");
_Static_assert(FLASH_DISK_BASE_OFFSET +
  FLASH_DISK_SLOT_COUNT * FLASH_DISK_SLOT_BYTES == PICO_FLASH_SIZE_BYTES,
  "disk slots must end at physical flash boundary");
_Static_assert(FLASH_JOURNAL_PAIR_COUNT == 8,
  "journal layout no longer matches the partition table");
_Static_assert(PICO_FLASH_ASSUME_CORE1_SAFE,
  "core 0 journal recovery runs only before core 1 is launched");

static const uint8_t *flash_disk_slot_ptr(unsigned drive) {
  uint32_t offset = FLASH_DISK_BASE_OFFSET + drive * FLASH_DISK_SLOT_BYTES;
  return (const uint8_t *)(XIP_BASE + offset);
}

static uint32_t crc32_bytes(const uint8_t *data, size_t length) {
  uint32_t crc = UINT32_MAX;
  for (size_t i = 0; i < length; ++i) {
    crc ^= data[i];
    for (unsigned int bit = 0; bit < 8; ++bit)
      crc = (crc >> 1) ^ (0xEDB88320u & (uint32_t)-(int32_t)(crc & 1u));
  }
  return ~crc;
}

enum { Z80_BOOT_MAGIC = 0x5442385Au, Z80_BOOT_VERSION = 1u }; // "Z8BT".

typedef struct {
  uint32_t magic;
  uint16_t version;
  uint16_t header_bytes;
  uint32_t image_bytes;
  uint32_t image_crc32;
  uint32_t header_crc32;
} z80_boot_manifest_t;

_Static_assert(sizeof(z80_boot_manifest_t) == 20,
  "boot manifest layout must match the host packer");

// Core 0 only, entirely synchronous: a flash read needs no filesystem,
// no queue, and no core 1 task, unlike the SD design this replaces.
```

```
static bool prepare_reset_held_dma(void) {
  isolate_buses();
  gpio_put(PIN_RESET_N, 0);
  for (unsigned int cycle = 0; cycle < 3; ++cycle)
    clock_one_cycle(1); // Bit-banged SIO pulses; set_z80_clock_hz() runs later.
  stop_z80_clock();

  if (!gpio_get(PIN_BUSACK_N) || !gpio_get(PIN_IORQ_N) ||
      !gpio_get(PIN_RD_N) || !gpio_get(PIN_WR_N))
    return false;

  return true;  // RESET# LOW selects the Pico SRAM controls (Section 1.2).
}

static bool boot_image_from_flash(void) {
  const z80_boot_manifest_t *manifest =
    (const z80_boot_manifest_t *)(XIP_BASE + FLASH_BOOT_BASE_OFFSET);
  const uint8_t *source =
    (const uint8_t *)(XIP_BASE + FLASH_BOOT_PAYLOAD_OFFSET);

  if (manifest->magic != Z80_BOOT_MAGIC ||
      manifest->version != Z80_BOOT_VERSION ||
      manifest->header_bytes != sizeof(*manifest) ||
      manifest->image_bytes == 0 || manifest->image_bytes > SRAM_SIZE_BYTES)
    return false;

  uint32_t header_crc = crc32_bytes((const uint8_t *)manifest,
    offsetof(z80_boot_manifest_t, header_crc32));
  if (header_crc != manifest->header_crc32 ||
      crc32_bytes(source, manifest->image_bytes) != manifest->image_crc32)
    return false;

  if (!prepare_reset_held_dma())
    return false;

  for (uint32_t i = 0; i < manifest->image_bytes; ++i)
    dma_write_byte((uint16_t)i, source[i]);

  bool ok = true;
  for (uint32_t i = 0; i < manifest->image_bytes; ++i) {
    if (dma_read_byte((uint16_t)i) != source[i]) {
      printf("boot verify failed near %04lx\n", (unsigned long)i);
      ok = false;
      break;
    }
  }

  if (ok)
    printf("loaded boot bytes=%lu crc32=%08lx\n",
      (unsigned long)manifest->image_bytes,
      (unsigned long)manifest->image_crc32);
  isolate_buses();
  // RESET# remains asserted; the caller releases it only after success.
  return ok;
}

static bool flash_recover_journal(void);

static bool boot_cpm_from_flash(void) {
  return flash_recover_journal() && boot_image_from_flash();
}

enum { FLASH_SERVICE_ACQUIRE_BUS, FLASH_SERVICE_RELEASE_BUS };

typedef struct {
  uint8_t operation;
} flash_service_request_t;

static queue_t flash_service_request_queue;  // Core 1 -> Core 0.
static queue_t flash_service_result_queue;   // Core 0 -> Core 1.

static void flash_service_queues_init(void) {
  queue_init(&flash_service_request_queue, sizeof(flash_service_request_t), 4);
  queue_init(&flash_service_result_queue, sizeof(bool), 4);
}

// Core 0's foreground loop only; never called from io_trap_handler().
static void core0_service_flash_requests(void) {
  flash_service_request_t request;
  if (!queue_try_remove(&flash_service_request_queue, &request))
    return;
```

```
  bool ok;
  switch (request.operation) {
    case FLASH_SERVICE_ACQUIRE_BUS:
      ok = request_cpu_bus(500000);
      if (ok)
        disable_io_trap();              // BUSACK# LOW: no new cycle can start.
      break;
    case FLASH_SERVICE_RELEASE_BUS:
      enable_io_trap();                 // Arm before BUSACK# lets the Z80 run.
      ok = release_cpu_bus(500000);
      if (!ok)
        disable_io_trap();              // release_cpu_bus() asserted RESET#.
      break;
    default:
      ok = false;
      break;
  }
  queue_add_blocking(&flash_service_result_queue, &ok);
}

// Core 1 only; blocks its caller, never core 0's foreground loop.
static bool core0_request(uint8_t operation) {
  flash_service_request_t request = { operation };
  queue_add_blocking(&flash_service_request_queue, &request);
  bool ok = false;
  queue_remove_blocking(&flash_service_result_queue, &ok);
  return ok;
}

enum { FLASH_JOURNAL_MAGIC = 0x314C4E4Au }; // "JNL1".

typedef struct {
  uint32_t magic;
  uint32_t sequence;
  uint32_t target_offset;
  uint32_t data_crc32;
  uint32_t header_crc32;
  uint8_t reserved[FLASH_PAGE_SIZE - 20u];
} flash_journal_header_t;

_Static_assert(sizeof(flash_journal_header_t) == FLASH_PAGE_SIZE,
  "journal header must occupy one flash page");

typedef struct {
  uint32_t header_offset;
  uint32_t data_offset;
  uint32_t target_offset;
  uint8_t *data;
  flash_journal_header_t header;
  bool committed;
} flash_write_params_t;

static bool flash_disk_block_offset_valid(uint32_t offset) {
  uint32_t disk_end = FLASH_DISK_BASE_OFFSET +
    FLASH_DISK_SLOT_COUNT * FLASH_DISK_SLOT_BYTES;
  return offset >= FLASH_DISK_BASE_OFFSET &&
    offset <= disk_end - FLASH_SECTOR_SIZE &&
    (offset & (FLASH_SECTOR_SIZE - 1u)) == 0;
}

static _Noreturn void reboot_after_flash_safe_failure(void) {
  gpio_put(PIN_RESET_N, 0);
  isolate_buses();
  stop_z80_clock();
  watchdog_reboot(0, 0, 0);
  while (true)
    tight_loop_contents();
}

static void flash_write_callback(void *param) {
  flash_write_params_t *p = (flash_write_params_t *)param;
  flash_range_erase(p->header_offset, FLASH_JOURNAL_PAIR_BYTES);
  flash_range_program(p->data_offset, p->data, FLASH_SECTOR_SIZE);
  flash_range_program(p->header_offset, (const uint8_t *)&p->header,
    FLASH_PAGE_SIZE);
  flash_range_erase(p->target_offset, FLASH_SECTOR_SIZE);
  flash_range_program(p->target_offset, p->data, FLASH_SECTOR_SIZE);

  p->committed = memcmp((const void *)(XIP_BASE + p->target_offset),
    p->data, FLASH_SECTOR_SIZE) == 0;
  if (p->committed) {
    flash_range_erase(p->header_offset, FLASH_SECTOR_SIZE);
```

```
    p->committed = *(const uint32_t *)(XIP_BASE + p->header_offset) ==
      UINT32_MAX;
  }
}

typedef struct {
  uint32_t header_offset;
  uint32_t target_offset;
  uint8_t *data;
  bool restored;
} flash_restore_params_t;

static void flash_restore_callback(void *param) {
  flash_restore_params_t *p = (flash_restore_params_t *)param;
  flash_range_erase(p->target_offset, FLASH_SECTOR_SIZE);
  flash_range_program(p->target_offset, p->data, FLASH_SECTOR_SIZE);
  p->restored = memcmp((const void *)(XIP_BASE + p->target_offset),
    p->data, FLASH_SECTOR_SIZE) == 0;
  if (p->restored) {
    flash_range_erase(p->header_offset, FLASH_SECTOR_SIZE);
    p->restored = *(const uint32_t *)(XIP_BASE + p->header_offset) ==
      UINT32_MAX;
  }
}

// Core 0 only, before core 1 launch and before RESET# release.
static bool flash_recover_journal(void) {
  static flash_journal_header_t headers[FLASH_JOURNAL_PAIR_COUNT];
  static bool valid[FLASH_JOURNAL_PAIR_COUNT];
  static uint8_t recovery_block[FLASH_SECTOR_SIZE];

  for (unsigned pair = 0; pair < FLASH_JOURNAL_PAIR_COUNT; ++pair) {
    uint32_t header_offset = FLASH_JOURNAL_BASE_OFFSET +
      pair * FLASH_JOURNAL_PAIR_BYTES;
    memcpy(&headers[pair], (const void *)(XIP_BASE + header_offset),
      sizeof headers[pair]);
    valid[pair] = headers[pair].magic == FLASH_JOURNAL_MAGIC &&
      headers[pair].header_crc32 == crc32_bytes((const uint8_t *)&headers[pair],
        offsetof(flash_journal_header_t, header_crc32)) &&
      flash_disk_block_offset_valid(headers[pair].target_offset);
  }

  for (unsigned recovered = 0; recovered < FLASH_JOURNAL_PAIR_COUNT;
      ++recovered) {
    int selected = -1;
    for (unsigned pair = 0; pair < FLASH_JOURNAL_PAIR_COUNT; ++pair) {
      if (valid[pair] && (selected < 0 || headers[pair].sequence <
          headers[(unsigned)selected].sequence))
        selected = (int)pair;
    }
    if (selected < 0)
      break;

    unsigned pair = (unsigned)selected;
    uint32_t header_offset = FLASH_JOURNAL_BASE_OFFSET +
      pair * FLASH_JOURNAL_PAIR_BYTES;
    uint32_t data_offset = header_offset + FLASH_SECTOR_SIZE;
    memcpy(recovery_block, (const void *)(XIP_BASE + data_offset),
      sizeof recovery_block);
    if (crc32_bytes(recovery_block, sizeof recovery_block) !=
        headers[pair].data_crc32)
      return false;

    flash_restore_params_t params = {
      header_offset, headers[pair].target_offset, recovery_block, false
    };
    int rc = flash_safe_execute(flash_restore_callback, &params, 1000);
    if (rc != PICO_OK || !params.restored)
      return false;
    valid[pair] = false;
  }
  return true;
}

// Core 1 only. Rewrites one flash sector of a CP/M disk slot, freezing
// the Z80 for the whole operation so its PWM-driven clock never
// advances while flash_safe_execute() pauses XIP fetches on both cores
// (Section 6.3). Core 0 must call flash_safe_execute_core_init() before
// core 1 starts, because core 0 is the victim parked by this call.
static bool flash_disk_write_sector(unsigned drive, uint32_t sector_offset,
    const uint8_t *data, size_t length) {
  uint32_t within_block = sector_offset & (FLASH_SECTOR_SIZE - 1u);
```

```
  if (drive >= FLASH_DISK_SLOT_COUNT || data == NULL || length == 0 ||
      sector_offset >= FLASH_DISK_SLOT_BYTES ||
      length > FLASH_DISK_SLOT_BYTES - sector_offset ||
      length > FLASH_SECTOR_SIZE - within_block)
    return false;

  uint32_t block_offset = sector_offset - within_block;
  uint32_t flash_offset = FLASH_DISK_BASE_OFFSET +
    drive * FLASH_DISK_SLOT_BYTES + block_offset;

  static uint8_t block[FLASH_SECTOR_SIZE];
  memcpy(block, (const uint8_t *)(XIP_BASE + flash_offset), sizeof block);
  memcpy(block + within_block, data, length);

  if (!core0_request(FLASH_SERVICE_ACQUIRE_BUS))
    return false;

  static uint32_t sequence;
  static unsigned next_pair;
  unsigned pair = next_pair++ % FLASH_JOURNAL_PAIR_COUNT;
  uint32_t header_offset = FLASH_JOURNAL_BASE_OFFSET +
    pair * FLASH_JOURNAL_PAIR_BYTES;

  flash_write_params_t params;
  memset(&params, 0, sizeof params);
  params.header_offset = header_offset;
  params.data_offset = header_offset + FLASH_SECTOR_SIZE;
  params.target_offset = flash_offset;
  params.data = block;
  memset(&params.header, 0xFF, sizeof params.header);
  params.header.magic = FLASH_JOURNAL_MAGIC;
  params.header.sequence = ++sequence;
  params.header.target_offset = flash_offset;
  params.header.data_crc32 = crc32_bytes(block, sizeof block);
  params.header.header_crc32 = crc32_bytes((const uint8_t *)&params.header,
    offsetof(flash_journal_header_t, header_crc32));

  int rc = flash_safe_execute(flash_write_callback, &params, 1000);
  if (rc != PICO_OK)
    reboot_after_flash_safe_failure();
  bool released = core0_request(FLASH_SERVICE_RELEASE_BUS);

  return released && params.committed;
}

enum {
  DISK_COMMAND_STATUS_PORT = 0x10,
  DISK_DRIVE_PORT = 0x11,
  DISK_LBA_LOW_PORT = 0x12,
  DISK_LBA_HIGH_PORT = 0x13,
  DISK_DATA_PORT = 0x14,
  DISK_COMMAND_CLEAR = 0,
  DISK_COMMAND_READ = 1,
  DISK_COMMAND_WRITE = 2,
  DISK_STATUS_READY = 1u << 0,
  DISK_STATUS_DATA_READY = 1u << 1,
  DISK_STATUS_DATA_ROOM = 1u << 2,
  DISK_STATUS_BUSY = 1u << 3,
  DISK_STATUS_ERROR = 1u << 7,
  DISK_WRITE_QUEUE_DEPTH = 2
};

typedef struct {
  uint8_t drive;
  uint16_t lba;
  uint8_t data[FLASH_DISK_RECORD_BYTES];
} disk_write_request_t;

static queue_t disk_write_queue;             // Z80/Core 0 -> Core 1.
static uint32_t disk_status = DISK_STATUS_READY;
static uint32_t disk_fatal_error;
static uint8_t disk_drive;
static uint16_t disk_lba;
static uint8_t disk_write_drive;   // Snapshot of drive/lba at WRITE issue,
static uint16_t disk_write_lba;    // immune to changes during data transfer.
static uint16_t disk_data_index;
static uint8_t disk_data[FLASH_DISK_RECORD_BYTES];

static uint32_t disk_status_load(void) {
  return __atomic_load_n(&disk_status, __ATOMIC_ACQUIRE);
}
```

```
static void disk_status_store(uint32_t status) {
  __atomic_store_n(&disk_status, status, __ATOMIC_RELEASE);
}

static bool disk_address_valid(void) {
  return disk_drive < FLASH_DISK_SLOT_COUNT &&
    disk_lba < FLASH_DISK_RECORD_COUNT;
}

static void disk_service_init(void) {
  queue_init(&disk_write_queue, sizeof(disk_write_request_t),
    DISK_WRITE_QUEUE_DEPTH);
  __atomic_store_n(&disk_fatal_error, 0, __ATOMIC_RELEASE);
  disk_status_store(DISK_STATUS_READY);
}

static void disk_start_command(uint8_t command) {
  if (disk_status_load() & DISK_STATUS_BUSY)
    return;

  bool fatal = __atomic_load_n(&disk_fatal_error, __ATOMIC_ACQUIRE);
  if (command == DISK_COMMAND_CLEAR) {
    disk_data_index = 0;
    disk_status_store(DISK_STATUS_READY | (fatal ? DISK_STATUS_ERROR : 0));
    return;
  }
  if (fatal) {
    disk_status_store(DISK_STATUS_READY | DISK_STATUS_ERROR);
    return;
  }
  if (!disk_address_valid()) {
    disk_status_store(DISK_STATUS_READY | DISK_STATUS_ERROR);
    return;
  }

  disk_data_index = 0;
  if (command == DISK_COMMAND_READ) {
    const uint8_t *source = flash_disk_slot_ptr(disk_drive) +
      (uint32_t)disk_lba * FLASH_DISK_RECORD_BYTES;
    memcpy(disk_data, source, sizeof disk_data);
    disk_status_store(DISK_STATUS_READY | DISK_STATUS_DATA_READY);
  } else if (command == DISK_COMMAND_WRITE) {
    disk_write_drive = disk_drive;
    disk_write_lba = disk_lba;
    disk_status_store(DISK_STATUS_READY | DISK_STATUS_DATA_ROOM);
  } else {
    disk_status_store(DISK_STATUS_READY | DISK_STATUS_ERROR);
  }
}

static uint8_t disk_virtual_io_read(uint8_t port) {
  if (port == DISK_COMMAND_STATUS_PORT)
    return (uint8_t)disk_status_load();
  if (port != DISK_DATA_PORT ||
      !(disk_status_load() & DISK_STATUS_DATA_READY))
    return 0;

  uint8_t value = disk_data[disk_data_index++];
  if (disk_data_index == sizeof disk_data)
    disk_status_store(DISK_STATUS_READY);
  return value;
}

static void disk_virtual_io_write(uint8_t port, uint8_t value) {
  uint32_t status = disk_status_load();
  if (port == DISK_COMMAND_STATUS_PORT) {
    disk_start_command(value);
  } else if (status & DISK_STATUS_BUSY) {
    return;
  } else if (port == DISK_DRIVE_PORT) {
    disk_drive = value;
  } else if (port == DISK_LBA_LOW_PORT) {
    disk_lba = (uint16_t)((disk_lba & 0xFF00u) | value);
  } else if (port == DISK_LBA_HIGH_PORT) {
    disk_lba = (uint16_t)((disk_lba & 0x00FFu) | ((uint16_t)value << 8));
  } else if (port == DISK_DATA_PORT &&
      (status & DISK_STATUS_DATA_ROOM)) {
    disk_data[disk_data_index++] = value;
    if (disk_data_index == sizeof disk_data) {
      disk_write_request_t request = {
        .drive = disk_write_drive, .lba = disk_write_lba
      };
```

```
      memcpy(request.data, disk_data, sizeof request.data);
      if (queue_try_add(&disk_write_queue, &request))
        disk_status_store(DISK_STATUS_BUSY);
      else
        disk_status_store(DISK_STATUS_READY | DISK_STATUS_ERROR);
    }
  }
}

// Core 1 only; call on every foreground-loop iteration, including while
// Wi-Fi is disconnected.
static void core1_service_disk_write(void) {
  disk_write_request_t request;
  if (!queue_try_remove(&disk_write_queue, &request))
    return;

  uint32_t offset = (uint32_t)request.lba * FLASH_DISK_RECORD_BYTES;
  bool ok = flash_disk_write_sector(request.drive, offset,
    request.data, sizeof request.data);
  if (!ok)
    __atomic_store_n(&disk_fatal_error, 1, __ATOMIC_RELEASE);
  disk_status_store(DISK_STATUS_READY | (ok ? 0 : DISK_STATUS_ERROR));
}
```

The Z80 BIOS sets drive and 16-bit LBA, then writes command 1 for a read or command 2 for a write. A read returns 128 bytes from the data port; a write accepts exactly 128 bytes there, then changes status to BUSY until core 1 finishes the journaled update. Command 0 clears a transient protocol/queue error when not busy; a flash or journal failure remains latched until reboot and recovery. The BIOS must poll READY/DATA_READY/DATA_ROOM/BUSY instead of assuming host timing. No erase, program, blocking queue, or flash-safe call runs inside `io_trap_handler()`.

WebSocket Terminal I/O Bridge (Final Phase 10 Integration)

Maintained source: terminal.h, terminal_bridge.c, and terminal_network.cpp.

The terminal bridge follows the `pico-altair-8800` model: initialize the queues on core 0, load the boot image from flash (Section 6.3, entirely on core 0), then launch core 1 to own Wi-Fi and WebSocket work -- and, per Section 6.3, the flash disk-write service, in the same task. The exact HTTP/WebSocket library can be `pico-ws-server` as in that project, or another lwIP-based server with the same callback shape. Only the queue functions are visible to the Z80 trap.

```
#include "pico/time.h"
#include "pico/multicore.h"
#include "pico/util/queue.h"

enum {
  TERM_DATA_PORT = 0x00,
  TERM_STATUS_PORT = 0x01,
  TERM_RX_DEPTH = 128,
  TERM_TX_DEPTH = 512,
  TERM_STATUS_RX_READY = 1u << 0,
  TERM_STATUS_TX_ROOM = 1u << 1,
  TERM_STATUS_CLIENT = 1u << 7
};

static queue_t terminal_rx_queue;  // Browser/Core 1 -> Z80/Core 0.
static queue_t terminal_tx_queue;  // Z80/Core 0 -> Browser/Core 1.
static uint32_t terminal_client_connected;
static uint32_t terminal_rx_drop_count;
static uint32_t terminal_tx_drop_count;

static void terminal_queues_init(void) {
  queue_init(&terminal_rx_queue, sizeof(uint8_t), TERM_RX_DEPTH);
  queue_init(&terminal_tx_queue, sizeof(uint8_t), TERM_TX_DEPTH);
}

uint8_t process_virtual_io_read(uint8_t port) {
  if (port >= DISK_COMMAND_STATUS_PORT && port <= DISK_DATA_PORT)
    return disk_virtual_io_read(port);

  if (port == TERM_DATA_PORT) {
    uint8_t value = 0;
    queue_try_remove(&terminal_rx_queue, &value);
    return value;
  }
```

```
  if (port == TERM_STATUS_PORT) {
    uint8_t status = 0;
    if (queue_get_level(&terminal_rx_queue) != 0)
      status |= TERM_STATUS_RX_READY;
    if (queue_get_level(&terminal_tx_queue) < TERM_TX_DEPTH)
      status |= TERM_STATUS_TX_ROOM;
    if (__atomic_load_n(&terminal_client_connected, __ATOMIC_ACQUIRE))
      status |= TERM_STATUS_CLIENT;
    return status;
  }

  return 0xFF;
}

void process_virtual_io_write(uint8_t port, uint8_t value) {
  if (port >= DISK_COMMAND_STATUS_PORT && port <= DISK_DATA_PORT) {
    disk_virtual_io_write(port, value);
    return;
  }
  if (port != TERM_DATA_PORT)
    return;
  if (!queue_try_add(&terminal_tx_queue, &value))
    __atomic_fetch_add(&terminal_tx_drop_count, 1, __ATOMIC_RELAXED);
}

// Called by the WebSocket server on core 1 when browser bytes arrive.
static bool terminal_ws_receive(const uint8_t *payload, size_t length,
    void *user_data) {
  (void)user_data;
  for (size_t i = 0; i < length; ++i) {
    uint8_t value = payload[i] == '\n' ? '\r' : payload[i];
    if (!queue_try_add(&terminal_rx_queue, &value)) {
      uint8_t discard;
      queue_try_remove(&terminal_rx_queue, &discard);
      if (!queue_try_add(&terminal_rx_queue, &value))
        __atomic_fetch_add(&terminal_rx_drop_count, 1, __ATOMIC_RELAXED);
    }
  }
  return true;
}

// Called by the WebSocket server on core 1 when it can send browser data.
static size_t terminal_ws_supply(uint8_t *buffer, size_t max_length,
    void *user_data) {
  (void)user_data;
  size_t count = 0;
  while (count < max_length && queue_try_remove(&terminal_tx_queue,
      &buffer[count]))
    ++count;
  return count;
}

static void terminal_ws_connected(void *user_data) {
  (void)user_data;
  __atomic_store_n(&terminal_client_connected, 1, __ATOMIC_RELEASE);
}

static void terminal_ws_disconnected(void *user_data) {
  (void)user_data;
  __atomic_store_n(&terminal_client_connected, 0, __ATOMIC_RELEASE);
  uint8_t discard;
  while (queue_try_remove(&terminal_rx_queue, &discard)) {}
  while (queue_try_remove(&terminal_tx_queue, &discard)) {}
}

enum { WS_OUTPUT_TIMER_INTERVAL_MS = 20, WS_INPUT_TIMER_INTERVAL_MS = 10 };

static uint32_t pending_ws_output;
static uint32_t pending_ws_input;
static struct repeating_timer ws_output_timer;
static struct repeating_timer ws_input_timer;

static bool ws_output_timer_callback(struct repeating_timer *t) {
  (void)t;
  __atomic_store_n(&pending_ws_output, 1, __ATOMIC_RELEASE);
  return true;                      // Keep repeating.
}

static bool ws_input_timer_callback(struct repeating_timer *t) {
  (void)t;
  __atomic_store_n(&pending_ws_input, 1, __ATOMIC_RELEASE);
  return true;
```

```
}

bool wifi_service_poll(void);
void terminal_websocket_server_start(uint16_t port,
  bool (*receive)(const uint8_t *, size_t, void *),
  size_t (*supply)(uint8_t *, size_t, void *),
  void (*connected)(void *), void (*disconnected)(void *));
void terminal_websocket_server_poll_output(void);
void terminal_websocket_server_poll_input(void);
void supervisor_usb_poll_nonblocking(void);

// The single core 1 entry point: WebSocket terminal and the Section 6.3
// flash disk-write service share this one task, as
// `multicore_launch_core1()` only accepts one function. The boot image
// is already in SRAM by the time this runs (Section 6.3/8.10).
static void core1_main(void) {
  bool websocket_started = false;

  while (true) {
    core1_service_disk_write();

    bool network_ready = wifi_service_poll();
    if (network_ready && !websocket_started) {
      terminal_websocket_server_start(8088, terminal_ws_receive,
        terminal_ws_supply, terminal_ws_connected, terminal_ws_disconnected);
      websocket_started = true;
    }
    if (network_ready && websocket_started &&
        __atomic_exchange_n(&pending_ws_output, 0, __ATOMIC_ACQ_REL)) {
      terminal_websocket_server_poll_output();
    }
    if (network_ready && websocket_started &&
        __atomic_exchange_n(&pending_ws_input, 0, __ATOMIC_ACQ_REL)) {
      terminal_websocket_server_poll_input();
    }
    tight_loop_contents();
  }
}

static bool start_core1_services(void) {
  terminal_queues_init();          // Core 0 creates queues before launch.
  flash_service_queues_init();
  disk_service_init();
  if (!flash_safe_execute_core_init())
    return false;                  // Core 0 registers as lockout victim.
  if (!add_repeating_timer_ms(-WS_OUTPUT_TIMER_INTERVAL_MS,
      ws_output_timer_callback, NULL, &ws_output_timer))
    return false;
  if (!add_repeating_timer_ms(-WS_INPUT_TIMER_INTERVAL_MS,
      ws_input_timer_callback, NULL, &ws_input_timer)) {
    cancel_repeating_timer(&ws_output_timer);
    return false;
  }
  multicore_launch_core1(core1_main);
  return true;
}

static void supervisor_fail_closed(const char *reason) {
  gpio_put(PIN_RESET_N, 0);
  isolate_buses();
  stop_z80_clock();
  printf("supervisor halted: %s\n", reason);
  while (true)
    tight_loop_contents();
}

int main(void) {
  diagnostic_safe_startup();       // First GPIO action; RESET# stays LOW.
  stdio_init_all();
  mcp_spi_init();

  if (!boot_cpm_from_flash())
    supervisor_fail_closed("boot package or journal recovery failed");
  if (!start_core1_services())
    supervisor_fail_closed("flash lockout or timer initialization failed");
  if (!set_z80_clock_hz(1000000))
    supervisor_fail_closed("invalid Z80 clock configuration");

  enable_io_trap();                // Arm before the Z80 can issue I/O.
  gpio_put(PIN_RESET_N, 1);        // Boot image verified; begin execution.

  while (true) {
```

```
    core0_service_flash_requests();
    supervisor_usb_poll_nonblocking();
    tight_loop_contents();
  }
}
```

`wifi_service_poll()`, `terminal_websocket_server_start()`, and the two server poll functions stand for the network layer, not new Z80-facing logic. Their implementation belongs entirely to core 1 and should mirror the reference project's `core1_io_mgr.c` pattern. `wifi_service_poll()` is an idempotent, bounded lifecycle state machine: it initializes CYW43, enables station mode, and associates without sleeping; after a partial initialization failure it cleans up and retries with an internal backoff. Core 1 calls it on every loop even after the server starts. It returns false while unavailable, re-enters association after link loss, and returns true once the station link is usable again. This guarantees `core1_service_disk_write()` runs even with Wi-Fi absent or reconnecting. Once associated, disable power-saving with `cyw43_wifi_pm(&cyw43_state, CYW43_NO_POWERSAVE_MODE)` for lower terminal latency. `supervisor_usb_poll_nonblocking()` similarly stands for an optional command parser that must return promptly so core 0 cannot starve flash ownership requests.

Required Integration Order

Maintained source: Stage 10 main.c and Stage 10 CMakeLists.txt.

The command-loop application must call `diagnostic_safe_startup()` as its first GPIO action. After Phase 3 hardware is fitted, call `mcp_spi_init()` before any MCP access. Keep `enable_io_trap()` disabled during DMA and single-step operation; call `set_z80_clock_hz()` first, then enable the trap immediately before releasing RESET# for a PWM-run test. Before returning to DMA from a running CPU, call `disable_io_trap()` and either request the CPU bus while the clock still runs or assert RESET# and supply at least three further full clocks. Only after the selected ownership procedure completes and the Z80 bus is verified high-impedance may the firmware enable either transceiver. While the trap is enabled, reserve SPI0 for the handler: no other IRQ, core, or main-loop operation may access the MCP23S17. Drain `core0_service_flash_requests()` continuously from core 0's nonblocking foreground loop. For acquisition, leave trapping enabled while asserting BUSREQ# and waiting for BUSACK# LOW, then disable it. For release, enable trapping while BUSACK# is still LOW, then deassert BUSREQ# and wait for BUSACK# HIGH. Core 0 must call and check `flash_safe_execute_core_init()` before launching core 1 and must not use the multicore FIFO for anything else, because the lockout handler owns it. Journal recovery is the only core-0 flash write and occurs before core 1 launch; all runtime writes execute on core 1 while the Z80 is held in BUSACK#.

8.14 Local Datasheet Set

- Z84C00 CPU datasheet
- AS6C1008 SRAM datasheet
- MCP23017/MCP23S17 datasheet
- SN74AHCT125 datasheet
- SN74HCT245 datasheet
- SN74LVC245A datasheet
- SN74LVC244A datasheet
- SN74HCT157 datasheet
- SN74HCT08 datasheet
- 1N5817/1N5818/1N5819 Schottky diode datasheet
- RP2350 datasheet
- Raspberry Pi Pico 2 board datasheet
- BusBoard BB830 breadboard datasheet

8.15 Source Code Index

Canonical repository: github.com/gloveboxes/Z80ROMlessSBC. All phase applications are cumulative: each stage links the shared modules proven by earlier stages, while remaining independently buildable for hardware bring-up.

Project Build Files

File	Purpose
CMakeLists.txt	Pico SDK import, Pico 2 W target, project languages, and protected firmware flash linker boundary
src/CMakeLists.txt	Shared firmware libraries, stage registration, and the `z80_cpm_images` artifact target
.gitignore	Generated build, Python cache, and assembler-output exclusions
package.json, package-lock.json	Locked Mermaid and browser-automation dependencies plus the `npm run pdf` command
scripts/build-readme-pdf.mjs	Pandoc, Mermaid, and Chromium PDF renderer with full-page diagram layout

Cumulative Stage Applications

Phase	Application	Target definition	Responsibility
0	Power checklist	Hardware-only	Empty-socket wiring, rail, resistance, and startup-state checks
1	main.c	CMakeLists.txt	Safe supervisor startup and GPIO walk
2	main.c	CMakeLists.txt	Buffered controls and variable-frequency clock
3	main.c	CMakeLists.txt	MCP23S17 register and port diagnostics
4	main.c	CMakeLists.txt	16-bit address-bus drive, sample, and isolation tests
5	main.c	CMakeLists.txt	8-bit data-bus drive, sample, and isolation tests
6	main.c	CMakeLists.txt	SRAM DMA and full-memory validation
7	main.c	CMakeLists.txt	Z80 reset, execution, clock, and BUSREQ/BUSACK tests
8	main.c	CMakeLists.txt	Virtual-ROM preload and synchronous I/O trapping
9	main.c	CMakeLists.txt	Manifest boot, journal recovery, and persistent flash disks
10	main.c	CMakeLists.txt	Final CP/M, flash-disk, Wi-Fi, and WebSocket integration

Shared Firmware Modules

Module	Public interface	Implementation	Responsibility
Pin map	pins.h	Header-only	Authoritative Pico GPIO assignments
Supervisor	supervisor.h	supervisor.c	Safe GPIO startup and fail-safe isolation defaults
Clock	clock.h	clock.c	PWM frequency selection, stop/resume, and single-cycle clocking
MCP23S17	mcp23s17.h	mcp23s17.c	SPI register access and 16-bit address expansion
Buses	bus.h	bus.c	Contention-safe address/data direction, isolation, drive, and sample operations
SRAM	sram.h	sram.c	DMA byte access, image load/verify, and RAM diagnostics
CPU	cpu.h	cpu.c	Reset-held DMA, run control, bus ownership, and fail-closed state
I/O trap	io_trap.h	io_trap.c	Synchronous Z80 IN/OUT interception and fault counters
Flash disk	flash_disk.h	disk_device.c	Z80-facing command/status, drive, LBA, and 128-byte data ports
Flash layout	flash_layout.h	Header-only	Firmware, journal, boot-package, and disk-slot offsets
Flash backend	flash_backend.h	flash_backend.c	Manifest validation, SRAM boot load, journal recovery, and core-1 commits
Terminal	terminal.h	terminal_bridge.c, terminal_network.cpp	Queue-backed terminal ports, Wi-Fi lifecycle, HTTP, and WebSocket service

CP/M, Disk, and Web Tooling

Area	Files	Purpose
Native CP/M	README, bios.asm, build_images.py, test_build_images.py	Native BIOS, CCP/BDOS extraction, boot package, full-flash composition, and host regression tests
Disk geometry and conversion	README, geometry.py, convert_altair_disks.py	Shared CP/M geometry and deterministic Altair-media conversion
Generated disk artifacts	generated directory, manifest.json	Four exact 320 KiB intermediate disk slots and source/output hashes
Preserved source media	source-altair directory, altair_88dskrom.h, altair_disk_loader.h	Original framed disks, Altair loader references, and upstream license
Browser terminal	terminal.html, embed_html.py	Embedded Stage 10 terminal client and build-time HTML conversion
Network configuration	lwipopts.h, wifi_config.h.in	lwIP settings and build-time Wi-Fi credential template

Generate the PDF

The PDF renderer requires Node.js, Pandoc, and Microsoft Edge, Google Chrome, or Chromium. On macOS, install the command-line prerequisites with:

```
brew install node pandoc
```

From the repository root, install the locked dependencies and generate `README.pdf`:

```
npm ci
npm run pdf
```

The renderer preserves tables, code highlighting, local images, links, and MathML. Mermaid diagrams are emitted as vector graphics on dedicated pages; wide diagrams automatically use A4 landscape orientation.

Set `BROWSER_EXECUTABLE` if the browser is installed in a nonstandard location:

```
BROWSER_EXECUTABLE="/path/to/chrome" npm run pdf
```

To render another Markdown file or choose another output path, pass both paths after `--`:

```
npm run pdf -- README.md README.pdf
```